ARCHITECTURAL AND PATENT SPECIFICATION

TOWNSENDCAPSTONECORNER

Autopoietic self-referential LUT
Physical closure and determinism audit
Cross-domain signed-distance lattice

PHYSICAL AND GEOMETRIC CONSTRUCTION

TAU TAU MU MU = (τ_a, τ_c, μ_R, μ_P)
reaction kinetics + measurement and closure contracts
$S = (A \odot B) \odot C \rightarrow$ lattice → physical control → regeneration

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Inventor	Tom Klootwijk
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Contents

Physical exactness assessment 5

Architectural overview 6

1. Technical field and architectural objective 7

2. Spatial representation and field types 7

3. Construction of cross-domain signed-distance windows 9

4. Field lattice and double Hadamard coupling 11

5. Spatial lattice generator 13

6. Negative-solid construction and CSG difference 14

7. Computing structures and execution contract 16

8. Output geometry, feedback, and embodiments 17

9. Worked numerical constructions 19

10. Physical TAU TAU MU MU binding 20

11. Atomic reference and chemical clock realization 23

12. Coupled physical balances and the spatial lattice 25

13. Autopoietic self-referential lookup tables 27

14. Physical control, regeneration and operating cycles 30

15. Records, calibration and executable contracts 32

16. Worked clock-bound lattice construction 33

17. Apparatus embodiments and construction details 36

18. Description of drawings and reference numerals 38

19. Scope of physical exactness and evidence 40

20. Timebase, sensing and observable quantities 41

21. Thermodynamic and kinetic closure 42

22. Species transport, flow and changing volumes 44

23. Thermal, mechanical and structural realization 45

24. Initial/boundary conditions and deterministic solution 47

25. Stochastic chemistry and chemical-clock limits 49

26. Uncertainty, identifiability and independent validation 50

27. SDF, numerical geometry and LUT approximation 51

28. Control stability and physical regeneration 53

29. Quantitative exactness audit 54

30. Completion register and experimental decision criteria 56

Claims 59

Abstract 64

FIG. 1. SDF system architecture 65

FIG. 2. Typed field algebra 66

FIG. 3. SDF execution flow 67

FIG. 4. CSG lattice construction 68

FIG. 5. Cross-domain field slice 69

FIG. 6. Physical TAU TAU MU MU binding 70

FIG. 7. Physical apparatus and feedback 71

FIG. 8. Self-reference and regeneration 72

FIG. 9. Atomic and chemical clock binding 73

FIG. 10. Chemical progression and lattice geometry 74

FIG. 11. Chemical-clock conditioning and model mismatch 75

FIG. 12. Grid resolution and geometric uncertainty 76

Claim-support map and source record 77

Technical references 79

The description uses numbered paragraphs for stable reference. The drawing sheets and claim-support map follow the claims and abstract.

Physical exactness assessment

Version 3 answers two separate questions: whether the supplied mathematics and reconstruction are internally deterministic, and whether the architecture has been demonstrated to predict a real apparatus to a stated accuracy. The ideal reaction and retained LUT are deterministic within their declared models. Physical exactness of an operating apparatus has not been established by the supplied analytic execution records.

Equation 17A is a valid consequence of the chosen reversible reaction law, chemical-time inverse and detailed-balance relation. Its value of one is an algebraic identity when those same assumptions and parameters generate both sides. It is not an independent empirical confirmation of a universal four-quantity law. A physical comparison instead uses separately acquired observations and a calibration fixed before the validation interval.

Layer	Assessment	Current evidence
Defined constants and algebra	Exact at the level of the stated definitions and symbolic model.	SI definition, dimensional checks and a derivation.
Ideal R ⇌ P dynamics	Deterministic and uniquely solvable for fixed rates and initial state.	Analytic solution; domain and positivity checks.
LUT reconstruction	Deterministic for retained inputs and the specified implementation environment.	Six saved generations reconstruct exactly; 24 original tests pass.
Real chemical/atomic apparatus	Accuracy is unquantified by the supplied evidence.	No independent apparatus measurement set or calibrated error budget is supplied.
Physical lattice performance	Geometric construction is verified for the sampled model; strength, transport and fabrication accuracy require validation.	Generated mesh topology and volume checks; material and build records remain to be supplied.
Physical self-production	A resource and renewal architecture is specified; physical autonomy is not demonstrated.	Working-row regeneration is implemented; hardware renewal is an apparatus contract.

The $2.354383354941092 \times 10^{-11}$ s residual reported in version 2 is a generated count/analytic-model consistency result. It is not a measured clock accuracy of about 23.5 picoseconds. The 24 tests verify the supplied implementation; they do not establish the accuracy of unspecified reaction chemistry, sensing, materials or fabrication.

Sections 19-30 of the complete specification supply the closure equations and error audit. Each physical realization still requires its own identified substances, constitutive laws, calibrated coefficients and operating conditions; the equations alone do not establish universal physical exactness.

Architectural overview

TOWNSENDCAPSTONECORNER is an apparatus and architectural formalization joining an atomic time reference, a chemically evolving material system, an autopoietic self-referential lookup table, and the signed-distance-field cross-domain lattice of the preceding specification. Its physical binding is the ordered quantity tuple (τ_a, τ_c, μ_R, μ_P), named TAU TAU MU MU. The first pair describes atomically referenced elapsed time and time inferred from chemical progress. The second pair describes the chemical potentials of the reactant and product sides of a declared reaction.

The binding is realized through a clock observation, chemical measurement, reaction law, material balance, and energy balance. A measured state determines chemical time and affinity; those quantities constrain table regeneration and the spatial operating windows. The table generates lattice-control parameters and an actuation proposal. A physical realization adds a command receipt and a subsequent measurement of the chemical or structural result to close the loop.

Autopoiesis has an operational definition in this architecture: the system repeatedly produces and renews its own working table, predictive rows, dependency references, and geometry-control records from its retained organization and incoming physical observations. Self-reference is explicit: the table describes its own schema and regeneration rule and links each generation to its predecessor. The realization supplies the matter, energy, sensing, and actuation required to renew physical clock segments or structural components.

The SDF foundation is retained. Spatially registered biochemical and biomechanical windows are converted to bounded gates, coupled by $S = (A \odot B) \odot C$, and used to select or grade a spatial lattice. A negative solid is subtracted from a base solid using regularized constructive solid geometry. The normalized signed-product band is retained as a second coupling mode. Geometric threshold τ_g is dimensionless and is separate from both clock quantities.

Sections 1-9 retain the geometric construction and its reproducible numerical models. Sections 10-17 complete the physical binding, clock apparatus, regenerative table, actuation contract, and integrated worked construction. The four physical symbol definitions and their implementation details are supplied explicitly in this edition under the inventor's instruction to complete the missing structures. Sections 19-30 add the physical closure equations, stochastic and uncertainty models, numerical sensitivity audit and remaining empirical completion register. The physical accuracy verdict is stated before the overview.

> **INTEGRATED PHYSICAL AND GEOMETRIC CONSTRUCTION**
>
> $d\tau_c/d\tau_a = \{j_- / [c_T \lambda_{ref} (x_{eq}-x)]\}$
> $\times \{\exp[(\mu_R-\mu_P)/(RT)] - 1\}$
> $S = (A \odot B) \odot C; \quad \Omega_F = \mathrm{Reg}(\Omega_0 \setminus \Omega_V)$
> observed state → regenerative LUT → physical control

1. Technical field and architectural objective

[0001] The invention relates to computer-implemented volumetric design, implicit geometric modeling, constructive solid geometry, and the synthesis of spatial structures controlled by multiple physical domains. A principal embodiment couples biochemical and biomechanical operating windows to generate the internal architecture of a scaffold, interface, or other physical structural component.

[0002] The technical objective is to establish a reproducible mapping from spatial domain conditions to a material boundary. The mapping preserves a common coordinate system, distinguishes physical distance from dimensionless coupling values, and provides an explicit sequence from domain-window construction through lattice generation to solid subtraction and geometric export.

[0003] A biochemical window identifies a spatial operating region associated with a selected biochemical quantity or condition. A biomechanical window identifies a spatial operating region associated with a selected mechanical condition. A window is a spatial set and its field representation; it is independent of a graphical window in an operating system. Windows 3D Builder is an optional geometric authoring environment in the disclosed workflow.

[0004] The architecture comprises a registration layer (110), a signed-distance window layer (120), a gate mapping layer (130), a double Hadamard coupling layer (140), a lattice generator (150), a negative-solid constructor (160), a CSG difference evaluator (170), and an output layer (180). The output may also be returned to domain analysis modules to create a subsequent design iteration.

> **Architectural relation**
>
> domain windows → signed distance fields → bounded gates
> S = (A ⊙ B) ⊙ C → lattice control → negative solid
> final solid = base solid \ negative solid

[0005] The double Hadamard product denotes two successive elementwise multiplications over corresponding spatial samples. The field-to-geometry construction and the CSG operation give those multiplications their geometric role. The operators, operands, and physical meaning of each stage are defined below.

2. Spatial representation and field types

2.1 Common spatial frame

[0006] Let D be an evaluation domain in three-dimensional Euclidean space and let x be a physical position expressed in a declared length unit. A structured implementation samples D on a grid of shape (n_x, n_y, n_z), with origin o, positive spacing h = (h_x, h_y, h_z), and an orthonormal direction matrix R. The index order is (i, j, k) = (x, y, z). For cell-centered sampling, the index-to-world mapping is:

Equation 1

$$x_{ijk} = o + R\,\mathrm{diag}(h_x, h_y, h_z)\,[i+1/2, j+1/2, k+1/2]^T$$
$$0 \le i < n_x, \quad 0 \le j < n_y, \quad 0 \le k < n_z$$

[0007] All operands of a pointwise operation are evaluated at the same physical position. The registration layer compares shape, spacing, origin, orientation, length units, sample location convention, and domain coverage. Fields with different transforms are resampled explicitly onto the selected common grid. Scalar interpolation and binary-mask resampling use separately recorded rules. This distinction between index correspondence and physical correspondence also appears in the OpenVDB grid-combination interface [T1].

2.2 Typed fields

Symbol	Type / units	Meaning
φ_0	SDF / length	Signed distance to the base-solid boundary; negative in the base.
φ_A, φ_B	SDF / length	Distances to the biochemical and biomechanical window boundaries.
A, B	Gate / dimensionless	Window membership or transition weights in [0,1].
C	Gate or mask / dimensionless	Structural modulation or eligibility field in [0,1].
I, S	Scalar score / dimensionless	First product $I = A \odot B$; second product $S = I \odot C$.
f_L	Signed implicit field / length	Lattice-generating field, negative in the designated lattice region.
φ_L	SDF / length	Distance-calibrated form of the lattice field when required.
τ_g	Scalar / dimensionless	Selection threshold, with $0 < \tau_g \le 1$.
f_V, φ_V	Implicit field; SDF / length	Negative-solid field before and after distance reconstruction.
f_F, φ_F	Implicit field; SDF / length	Final-solid field before and after distance reconstruction.
P, V	Boolean fields	Protected-region membership and input-validity membership.
E, D_v, F	Boolean fields	Sampled base, removed cells, and retained cells.
φ_C	SDF / length	Structural boundary field in signed-product mode; distinct from gate C.

Symbol	Type / units	Meaning
P_s, η	Dimensionless	Normalized signed product and its magnitude-band parameter.
L_A, L_B, L_C	Positive lengths	Reference lengths normalizing the three signed-product operands.
d_Σ, b	Distance / length	Distance to the union of zero surfaces and selected physical half-width.

[0008] A signed distance field has magnitude equal to the Euclidean distance to a specified boundary and a sign indicating interior or exterior. A signed implicit field has the required sign and geometric boundary, while its magnitude may follow another scale. A gate and a score are dimensionless. These types are retained in the operation record so that a gate is never interpreted as a physical distance.

Equation 2: inside-negative signed distance

$$SD(\Omega)(x) = -\mathrm{dist}(x, \partial\Omega) \quad \text{for } x \text{ inside } \Omega$$
$$SD(\Omega)(x) = 0 \quad \text{for } x \text{ on } \partial\Omega$$
$$SD(\Omega)(x) = +\mathrm{dist}(x, \partial\Omega) \quad \text{for } x \text{ outside } \Omega$$

[0009] Rigid translations and rotations preserve distance magnitudes. A nonuniform deformation or rescaling can change those magnitudes. After such a transform, the representation is tagged as an implicit field until distance reconstruction supplies its Euclidean-distance form. Sparse or narrow-band storage additionally records the represented band and the interior/exterior background values.

3. Construction of cross-domain signed-distance windows

3.1 Operating sets

[0010] Let q_A(x) describe a selected biochemical quantity, such as concentration, concentration-gradient magnitude, transport demand, or a derived biochemical response. Let q_B(x) describe a selected scalar mechanical quantity. The system evaluates declared predicates g_A and g_B to obtain operating sets Ω_A and Ω_B. A predicate can be an interval, threshold, isosurface band, or a combination of domain conditions.

Equation 3

$$\Omega_A = \{x \in D : g_A(q_A(x), x) \text{ is true}\}$$
$$\Omega_B = \{x \in D : g_B(q_B(x), x) \text{ is true}\}$$
$$\varphi_A = SD(\Omega_A), \quad \varphi_B = SD(\Omega_B)$$

[0011] The operating-set definitions determine the meaning of interior. In a channel-removal embodiment, Ω_A identifies a biochemical demand region and Ω_B

identifies a region permitted by the selected mechanical design rule. A high mechanical gate therefore means that removal is permitted by that rule. If the input is stress, a low-stress operating set can be defined before distance reconstruction; raw stress magnitude is not used as an increasing removal gate.

[0012] If mechanical input is a tensor, a specified scalarization precedes the operating predicate. For example, an embodiment uses equivalent stress computed from the deviatoric stress tensor s. The tensor basis, physical units, load case, scalarization, and predicate parameters are part of the domain record. Multiple load cases may be combined by a declared maximum, envelope, or other explicit reduction.

Equation 4: an equivalent-stress embodiment

$$s = \sigma - \mathrm{trace}(\sigma)\, I_3 / 3$$
$$q_B = \mathrm{sqrt}((3/2)\, \Sigma_{ij}\, s_{ij}^2)$$
$$\Omega_B = \{x : q_B(x) \le \sigma_{limit}\}$$

[0013] A window can instead be supplied directly as an analytic or sampled SDF. In that case, the architecture records its sign convention and transform and uses it without reconstructing the original physical predicate. This supports imported simulation windows, analytic test windows, geometrically specified interfaces, and directly authored domain regions.

3.2 Gate maps

[0014] A gate map converts a distance field into a bounded coupling operand. A hard gate represents membership. A transition gate spreads the change across a prescribed physical width. For ε greater than zero, the linear transition shown below equals one at distances at least ε inside the window, one half at its boundary, and zero at distances at least ε outside the window.

Equation 5

$$w_{hard}(\varphi) = 1 \text{ if } \varphi \le 0; \text{ otherwise } 0$$
$$w_\varepsilon(\varphi) = \mathrm{clip}(1/2 - \varphi/(2\varepsilon), 0, 1), \quad \varepsilon > 0$$
$$A(x) = w_A(\varphi_A(x)), \qquad B(x) = w_B(\varphi_B(x))$$

[0015] An alternative smooth transition uses $u = \mathrm{clip}((\varphi + \varepsilon)/(2\varepsilon), 0, 1)$ and $w(\varphi) = 1 - 3u^2 + 2u^3$. Gate selection and transition widths are recorded independently for each domain. A hard gate is evaluated directly and does not require division by a zero transition width.

[0016] For a symmetric transition map, the transition zone is part of the blended operating rule and may extend outside the zero boundary. If an operating set is to impose strict exclusion, the implementation also applies its hard membership mask to C, or uses an interior-only gate such as $\mathrm{clip}(-\varphi/\varepsilon, 0, 1)$. The selected gate rule therefore explicitly determines whether transition-zone participation or strict set membership is required.

[0017] The structural field C specifies binary eligibility or continuous modulation. In a through-channel implementation, C remains active over the base and a surrounding

evaluation collar so the negative solid can extend through the base boundary. The final base boundary is enforced by the CSG difference. An explicit protected-region field can separately preserve support interfaces, attachment zones, or selected load paths.

4. Field lattice and double Hadamard coupling

4.1 Pointwise field lattice

[0018] Let $\mathcal{F}$ be the set of real-valued signed fields defined on the common evaluation domain. The pointwise order $f \preccurlyeq g$ means $f(x) \le g(x)$ at every position. The pointwise minimum is the meet and the pointwise maximum is the join. Under the inside-negative convention, these operators correspond respectively to geometric union and intersection of the represented interiors.

Equation 6

$$f \preccurlyeq g \iff f(x) \le g(x) \text{ for every } x$$
$$(f \wedge g)(x) = \min(f(x), g(x))$$
$$(f \vee g)(x) = \max(f(x), g(x))$$
union: min(f,g) intersection: max(f,g)
difference: max(f,-g)

[0019] The pointwise order is a field order; it does not imply that the numerically greater field represents the greater solid. The inside-negative convention reverses the direction of set inclusion associated with increasing field values. Boundary-only contacts and coincident surfaces are handled through the regularized-solid construction defined in Section 6.

[0020] Minimum and maximum supply Boolean sign composition. They are not, in general, a global reconstruction of exact distance to the resulting boundary. The distance-calibrated representation is obtained by applying SD to the composed solid when a subsequent operation uses metric distances. The distinction between geometric set composition and distance reconstruction is also treated in the distance-based modeling literature [T2].

4.2 Cross-domain coupling

[0021] At every registered sample, the coupling layer evaluates two ordinary Hadamard products in a fixed order. The first combines the biochemical and biomechanical gates. The second applies the structural modulation field. The resulting score is scalar and dimensionless.

Equation 7: double Hadamard product

$$I = A \odot B$$
$$S = I \odot C$$
$$S_{ijk} = (A_{ijk} B_{ijk}) C_{ijk}, \quad 0 \le S_{ijk} \le 1$$

[0022] For nonnegative operands no larger than one, S is no larger than any of A, B, and C at the same position. A zero factor suppresses the score. With hard gates,

multiplication implements the conjunction of the three memberships. With transition gates, it supplies a continuous coupling whose spatial distribution depends on all three factors.

[0023] The signed distance fields themselves remain geometric operands. Their raw product is not used as a substitute for an intersection: multiplication of signed values encodes sign parity, so different interior/exterior combinations can produce the same sign. Mapping the domain fields to nonnegative gates makes the coupling rule explicit while minimum and maximum retain the geometric Boolean semantics.

[0024] The two fixed-channel products require two multiplications per sample and O(N) arithmetic for N grid samples. Registration, physical-field analysis, lattice evaluation, distance reconstruction, surface extraction, and solid evaluation are separate stages with their own costs. A fused implementation can omit storage of I while retaining the declared evaluation order in its operation record.

[0025] The architecture uses S in a selection embodiment, a parameter-modulation embodiment, or both. Selection chooses where lattice geometry is active. Parameter modulation changes a specified local lattice property such as radius or sheet thickness. The mode is recorded because increasing S can affect topology differently in those embodiments.

4.3 Signed-product surface-lattice embodiment

[0026] A second coupling mode operates directly on signed geometry fields. Three signed fields for cross-domain boundaries are normalized by positive reference lengths, and two Hadamard products form a dimensionless signed product P_s. The selected region is a band of that product, so all constituent zero surfaces contribute to the surface lattice. This mode explicitly uses the magnitude of the product rather than treating its sign as an intersection.

> **Equation 7A: signed-product band**
>
> $$P_s = ((\varphi_A/L_A) \odot (\varphi_B/L_B)) \odot (\varphi_C/L_C)$$
> $$L_A > 0,\ L_B > 0,\ L_C > 0$$
> $$\Omega_{band} = \mathrm{Reg}(\{x : |P_s(x)| \le \eta\}), \quad \eta > 0$$

[0027] The third signed field φ_C in this mode represents a structural boundary family and is distinct from the dimensionless gate C in Equation 7. The normalized product vanishes on the union of the constituent zero surfaces. A product-value band has width governed by the local product variation. For a prescribed physical half-width b, a distance-to-surface-union construction instead uses $d_\Sigma = \min(|\varphi_A|, |\varphi_B|, |\varphi_C|)$ when all three inputs are exact boundary-distance fields, and selects $d_\Sigma \le b$.

> **Equation 7B: combined product and physical-distance band**
>
> $$d_\Sigma = \min(|\varphi_A|, |\varphi_B|, |\varphi_C|)$$
> $$\Omega_{band,combined} = \mathrm{Reg}(\{|P_s| \le \eta\} \cap \{d_\Sigma \le b\}), \quad b > 0$$

[0028] The surface-lattice band is used directly as a cutter or intersected with another lattice or operating-region field. The base-minus-cutter operation and output contract

remain the same. The CouplingSpec records either window-gate selection or signed-product band selection, including all reference lengths, band parameters, and distance-reconstruction choices.

5. Spatial lattice generator

5.1 Repeated primitive network

[0029] A spatial lattice is specified by a cell transform, a set of primitive generators, and a repetition or placement rule. Let H contain three independent cell vectors and let n be an integer lattice index. A primitive field f_j placed at Hn is combined with the other placed primitives by a pointwise minimum. A bounded evaluation domain limits the repeated family to the portion needed for the structural envelope and its evaluation collar.

Equation 8

$\Lambda = \{Hn : n \in \mathbb{Z}^3\}$

$f_L(x) = \min \text{ over } (n,j) \text{ of } f_j(x - Hn; \theta_j)$

[0030] For a channel lattice, negative values designate the channel volume to be removed. For a strut lattice, negative values designate a solid network; the adapter explicitly converts that network to a retained-region or complementary cutter representation before the final difference. The interpretation is carried by the lattice record and is not inferred from a filename.

5.2 Parallel and connected channel embodiments

[0031] A parallel cylindrical channel family along the z direction has centers (a_m, b_n) and channel radii r_mn. The field below is an exact cylinder distance for each channel primitive. A minimum over the primitives gives the union's signed implicit representation. A connected three-dimensional network can add analogous families along x and y or connect channel centerlines with capsule primitives.

Equation 9

$f_z(x,y,z) = \min \text{ over } (m,n) [\sqrt{((x-a_m)^2 + (y-b_n)^2)} - r_{mn}]$

$f_L = \min(f_x, f_y, f_z)$ for a three-family channel network

[0032] For a capsule with centerline endpoints a and b and radius r, the closest centerline parameter is $t = \text{clip}(((x-a)\cdot(b-a))/||b-a||^2, 0, 1)$, and the primitive field is $||x-(a+t(b-a))||-r$. Nonzero segment length is required. This representation supports a finite edge network whose edge positions, directions, and radii are stored explicitly.

5.3 Grading and sheet embodiments

[0033] In a graded-radius embodiment, the coupling score controls a local radius between declared bounds. Higher score enlarges the channel radius for the removal convention used here. A gradient limiter or a cellwise parameter interpolation can constrain variation. The radius map is expressed in the same length unit as the lattice coordinates.

Equation 10

$$r(x) = r_{min} + (r_{max} - r_{min})\, S(x)$$
$$0 < r_{min} \le r_{max}$$
$$f_L(x) = d_{axis}(x) - r(x)$$

[0034] A varying-radius expression is tagged as a signed implicit field. If a downstream step requires a metric offset or a physical-distance query, SD reconstructs the boundary distance. A sheet lattice can be generated around a distance-calibrated mid-surface by $f_{sheet} = |\varphi_{mid}| - t(x)/2$, where t is the requested local sheet thickness. Repetition and spatial grading can be combined in the same generator.

[0035] The generator records cell vectors or channel centers, primitive type, radius or thickness parameters, placement extent, modulation mode, and field type. These structures make the lattice independently reproducible from the coupled field and allow another geometry engine to recreate the same construction.

6. Negative-solid construction and CSG difference

6.1 Set semantics

[0036] Let Reg(X) denote closure of the interior of X. A regularized solid retains volumetric material and its boundary while discarding isolated lower-dimensional remnants. Let Ω_0 be the base solid, Ω_L the designated negative lattice region, and Ω_P a protected region. The score threshold includes equality, as in the supplied disclosure.

Equation 11

$$\Omega_{sel} = \{x \in D : S(x) \ge \tau_g\}, \qquad 0 < \tau_g \le 1$$
$$\Omega_V = \mathrm{Reg}((\Omega_L \cap \Omega_{sel}) \setminus \Omega_P)$$
$$\Omega_F = \mathrm{Reg}(\Omega_0 \setminus \Omega_V)$$

[0037] The negative solid Ω_V is a cutter. It can extend outside Ω_0, particularly where a channel passes through the outer surface. The result remains bounded by Ω_0 because the base is the minuend of the final Boolean difference. The envelope collar and cutter extent are therefore distinct from the final material envelope.

6.2 Signed-field evaluation

[0038] Where the selected threshold is represented by a nondegenerate surface, a length-scaled score field f_sel = ℓ_sel(τ_g-S), with ℓ_sel greater than zero, supplies the correct interior sign. The scale ℓ_sel converts units and does not turn the score into a Euclidean distance. A protected-region field φ_P is negative inside the protected solid.

Equation 12: implicit construction

f_sel = ℓ_sel (τ_g - S)
f_V = max(f_L, f_sel, -φ_P) when protection is present
φ_V = SD(Ω_V)
f_F = max(φ_0, -φ_V)
φ_F = SD(Ω_F) when distance-calibrated output is required

[0039] The final maximum is evaluated against the negative-solid field, before the cutter is unnecessarily clipped to the base surface. When a mesh backend receives an already clipped negative solid, it evaluates the regularized set difference directly. This preserves through-openings without treating coincident zero-valued surface fragments as retained material. Empty-cutter and empty-result branches are evaluated before invoking boundary-distance reconstruction.

[0040] If S equals τ_g across a region with positive volume, f_sel has a zero plateau rather than an isolated boundary. In that case the inclusive membership rule in Equation 11 is authoritative: the selected set is constructed first and its boundary is then distance-reconstructed. A raw zero-isosurface extraction from the plateau is not the geometry definition. The same membership-first rule applies when sampled values fall exactly on a lattice boundary.

6.3 Discrete solid embodiment

[0041] In a cell-union implementation, E indicates cells belonging to the base, V indicates samples with valid domain values, P indicates protected cells, and L indicates lattice membership. The selected removal and retained material masks are evaluated by exact Boolean operations. Each selected cell denotes its physical cell volume rather than an interpolated surface sample.

Equation 13

D_v = E AND V AND NOT P AND L AND (S ≥ τ_g)
F = E AND NOT D_v
Ω_V,grid = Reg(union of physical cells with D_v = 1)
Ω_F,grid = Reg(union of physical cells with F = 1)

[0042] The cell-union surface is obtained from faces separating occupied and unoccupied cells. Internal shared faces are omitted. Each exposed quadrilateral is triangulated with outward orientation. Face adjacency, vertex links, edge incidence, connected components, and signed volume are evaluated on the resulting boundary. The cell-union output exactly represents the selected cells; it is a spatial discretization of the continuous SDF model.

[0043] A continuous surface-extraction implementation instead records sample placement, interpolation method, isovalue, boundary padding, and geometric tolerance. It extracts the boundary of the composed implicit field or reconstructed SDF. Any smoothing or remeshing is followed by reapplication of the base and protection constraints and by the output checks specified in Section 8.

7. Computing structures and execution contract

7.1 Data records

[0044] The architecture can be implemented on one processing system or divided among domain-analysis and geometry services. Each module accepts typed records and returns its result with operation metadata. The following structures supply the missing interfaces between the physical-domain windows, the field algebra, and the geometry engine.

Record	Required contents
GridSpec	Shape; axis order; cell or vertex sample convention; origin; spacing; direction matrix; length unit; evaluation bounds.
DomainWindow	Domain label; raw-data units and meaning or supplied-SDF identity; predicate and scalarization; field values; grid identity; sign convention; validity mask.
FieldRecord	Value array or analytic evaluator; type (SDF, implicit, gate, score, mask); unit; transform; background/band policy; operation provenance.
CouplingSpec	Mode; gate maps, widths, C and threshold for gate mode; φ_C identity, reference lengths L_A/L_B/L_C, η and optional b for signed-product mode; evaluation order, comparison rule, and numeric type.
LatticeSpec	Primitive family; cell transform or channel-center list; region meaning; placement extent; radius/thickness parameters; score-control mode.
BooleanNode	Operation name; ordered child references; represented field/solid type; protection inputs; distance-reconstruction policy.
GeometryResult	Retained solid or mesh; length unit; bounds; representation mode; topology/volume records; export format; build parameters.
IterationRecord	Input geometry version; domain-field versions; score parameters; output geometry; objective or feasibility values; termination reason.

7.2 Module operations

[0045] The registration operation produces a shared GridSpec and registered DomainWindow records. The window operation either constructs SDFs from operating predicates or accepts supplied SDFs with the declared convention. The gate operation maps those fields into A and B and constructs C. The coupling operation evaluates I and S. The lattice operation evaluates f_L using its recorded parameters and any score modulation.

[0046] The negative-solid operation combines lattice membership, threshold membership, and protected-region exclusion. The difference operation takes the base as its first operand and the negative solid as its second operand. The surface operation converts the retained representation into a boundary mesh. The output operation serializes the geometry and the metadata needed to reconstruct the model.

```
register inputs on a common spatial frame
construct or load phi_A, phi_B, and phi_base
A = gate_A(phi_A); B = gate_B(phi_B)
I = A * B
S = I * C
lattice = evaluate_lattice(parameters, S)
void = regularize(lattice AND (S >= tau) AND NOT protected)
final = regularized_difference(base, void)
if metric output requested: final_sdf = signed_distance(final)
extract boundary; evaluate output checks; export geometry and record
```

[0047] A dense implementation computes all samples. A sparse implementation computes active cells and carries explicit background membership. When a zero background score is used, the positive threshold prevents that background from being selected. A cropped field includes the exterior collar required by its cutter and surface-reconstruction operations.

7.3 Deterministic input handling

[0048] The execution contract rejects nonfinite values, inconsistent shapes, unregistered transforms, invalid gate widths, thresholds outside the declared range, and invalid lattice dimensions. A validity-mask embodiment excludes cells whose domain information is unavailable. It does not assign a missing value a meaning by silently replacing it with a domain score.

[0049] The comparison $S \geq \tau_g$ is evaluated in the declared numeric type and order. An optional boundary tolerance is an explicit parameter and is used consistently by selection and geometry reconstruction. Products are mathematically associative over the real numbers; a reproducible implementation nevertheless records its floating-point order. An empty cutter returns the base; full removal produces an empty-solid result record; the exporter handles each result explicitly.

8. Output geometry, feedback, and embodiments

8.1 Geometric output contract

[0050] The output layer records material volume, cutter volume within the base, bounding box, component count, boundary-edge incidence, vertex-link topology, orientation, and the coordinates and units of the exported geometry. It also records containment and preservation of designated protected regions. These values make the geometric result inspectable without repeating the domain-field calculations.

[0051] Additional design constraints can specify minimum material thickness, channel diameter, channel connectivity, accessibility of selected pore regions, or required attachment paths. These constraints use geometric distances, topology, or a declared analysis model. When distances are queried, the implementation uses the boundary geometry or a distance-calibrated field. An operation that changes topology or shape regenerates the corresponding check record.

[0052] The geometry adapter can emit an explicit mesh, a sampled signed field, or an operation tree. A Windows 3D Builder workflow imports a base model and the generated negative model, selects the base and cutter in the required order, executes the subtraction, and exports the resulting geometry. A scripted CSG workflow performs the same ordered operation through its chosen engine.

[0053] A core mesh 3MF output serializes evaluated geometry. An implementation can separately target a compatible Boolean-expression extension or toolpath extension. The current 3MF specification family defines these as distinct contracts [T3, T4]. The supplied implementation package uses STL geometry together with explicit units in JSON and an OpenSCAD difference program.

8.2 Geometry-driven iteration

[0054] In an iterative embodiment, the current retained geometry is returned to the biochemical and biomechanical analysis modules. They evaluate updated operating fields for that geometry. The window, coupling, lattice, and subtraction stages then generate another result using updated parameters or fields. The iteration record stores the geometry associated with each field set, avoiding the mixing of values from different geometry versions.

[0055] A declared objective may combine mechanical response, transport response, material volume, and geometric penalties with recorded scaling and weights. Alternatively, the system enforces a set of feasibility conditions and stops at the first result that meets them. The recorded stopping rule specifies how the objective or feasibility condition, geometric-change tolerance, and iteration budget determine termination.

> **Equation 14: one recorded objective form**
>
> $J(G) = \alpha\, J_{mechanical}(G) + \beta\, J_{transport}(G) + \gamma\, J_{geometry}(G)$
> termination: declared conditions met, geometric change small,
> or the recorded iteration budget is reached

8.3 Application embodiments

[0056] In a biochemical/biomechanical scaffold embodiment, the biochemical operating set indicates where a selected transport or concentration condition calls for channel access, and the mechanical operating set indicates the region permitted by the selected mechanical rule. Their SDF-derived gates jointly modulate the negative lattice. Protected attachment regions remain outside the cutter.

[0057] In an interface embodiment, signed windows describe bands around two domain boundaries. The product localizes the overlap of those bands, and the lattice generator

forms a spatial transition between connected structural regions. The transition width is specified in physical distance and can differ for each domain.

[0058] In a generalized cross-domain embodiment, additional registered domain gates can be introduced through further declared coupling stages. The principal double Hadamard embodiment retains the three operands A, B, and C. The architecture can use a nonperiodic centerline graph or a graded sheet family while preserving the same typed-field, coupling, and CSG interfaces.

[0059] A fabrication embodiment passes the evaluated solid to a geometry-compatible slicer, records material and process parameters, and generates the apparatus-specific build instructions. The physical manufacturing step is downstream of geometric evaluation and uses the recorded geometry version.

9. Worked numerical constructions

9.1 Exact coupling arithmetic

[0060] For a binary-exact arithmetic example, let $\tau_g = 0.5$ and assume lattice membership, valid input, and no protection at each listed sample. The values 0.5, 0.75, and 1.0 are exactly representable in binary floating point. The table therefore includes a direct equality case without decimal-representation ambiguity.

Case	A	B	C	S	Selection
Coupled support	0.75	0.75	1	0.5625	Removed
Mechanical restriction	0.75	0.25	1	0.1875	Retained
Envelope suppression	0.75	0.75	0	0	Retained
Threshold equality	1	0.5	1	0.5	Removed

[0061] A second arithmetic construction uses an $8 \times 8 \times 6$ cell base at 1 mm spacing, using a selection threshold of 0.72 and a selected 2×2 channel extending through all six z layers. The base contains 384 cells. The channel removes 24 cells and retains 360 cells, giving a removed fraction of 6.25 percent. Additional protected, invalid, and ineligible samples outside the selected through-channel exercise the separate selection gates without changing the channel count.

9.2 SDF lattice construction

[0062] The SDF example starts with a box distance field, a repeated cylindrical channel family, and analytic biochemical and biomechanical window fields. It evaluates the window gates, the two products, the lattice-membership condition, and the final Boolean membership on a declared grid. Its input fields are generated from the equations and configuration included in the package.

SDF construction parameter / result	Recorded value
Base dimensions	12 × 12 × 8 mm
Sampling grid	24 × 24 × 16 cells; 0.5 mm spacing
Channel families	Orthogonal x, y, z cylinders; 4 mm pitch
Channel radius / score threshold	0.8 mm / 0.55
Window boundaries	x = 10 mm; y = 9 mm
Gate transition half-widths	3 mm; 4 mm
Base cells / volume	9,216 / 1152 mm³
Unmodulated lattice cells / volume	3,600 / 450 mm³
Removed cells / volume	1,828 / 228.5 mm³
Retained cells / volume	7,388 / 923.5 mm³
Void / retained connected components	1 / 1
External CSG evaluation	OpenSCAD; successful

[0063] The exported cell-union mesh is assembled from exposed cell faces. The package also contains the CSG program used to subtract the exported negative solid from the exported base. The execution record identifies which Boolean evaluations were run and compares their geometric volumes with the occupied-cell volume.

[0064] The implementation checks the coupling order, monotonic stress-to-permission mapping where that mapping is used, threshold equality, protected and invalid samples, registration requirements, field-type semantics, signed-field subtraction, edge incidence, vertex links, connected components, and volume conservation for the example geometries. The numerical results and test output are supplied as machine-readable records.

10. Physical TAU TAU MU MU binding

10.1 Four quantities with physical referents

[0065] TAU TAU MU MU denotes four jointly bound physical quantities. It is a quantity relation, and each occurrence carries a physical location, an observation epoch, a unit, and a calibration or constitutive-model reference. The ordered tuple is $B_{phys} = (\tau_a, \tau_c, \mu_R, \mu_P)$. The two time quantities use seconds. The two potential quantities use joules per mole of reaction as fixed by the written stoichiometry. For a positive rescaling $\nu' = a\nu$, the reaction extent becomes $\xi' = \xi/a$ and each stoichiometric potential sum becomes a times its prior value, so $A_{r'}\, d\xi' = A_r\, d\xi$.

Quantity	Physical definition	Unit
τ_a	Elapsed time obtained from the atomic reference and its time-transfer record.	s
τ_c	Elapsed chemical time inferred from a measured reaction extent or unwrapped chemical phase using a retained calibration.	s
μ_R	Sum of reactant chemical potentials weighted by their stoichiometric coefficients.	J mol^{-1}
μ_P	Sum of product chemical potentials weighted by their stoichiometric coefficients.	J mol^{-1}
τ_g	Retained SDF score-selection threshold; legacy implementation parameter named tau.	1
$A_r = \mu_R - \mu_P$	Reaction affinity; distinct from the dimensionless biochemical gate A.	J mol^{-1}

Equation 15: the physical binding

$$B_{phys}(x,e) = (\tau_a(e), \tau_c(x,e), \mu_R(x,e), \mu_P(x,e))$$
$$\mu_R = \Sigma_i \alpha_i \mu_i, \quad \mu_P = \Sigma_i \beta_i \mu_i$$
$$A_r = \mu_R - \mu_P = -\Delta_r G$$
$$\Sigma_i \alpha_i X_i \rightleftharpoons \Sigma_i \beta_i X_i$$

[0066] For a species in a specified phase, μ_i is the partial molar Gibbs energy. A constitutive activity model writes $\mu_i = \mu_i^\circ(T,p) + RT \ln(a_i)$, where a_i is dimensionless. Standard state, temperature, pressure, activity convention, and reaction coefficients are recorded. IUPAC provides the underlying chemical-potential and affinity definitions [T6, T7]. Absolute reference choices are retained; reaction driving force uses the properly balanced difference $\mu_R - \mu_P$.

10.2 The exact relation supplied by a declared reaction

[0067] A principal physical embodiment uses the elementary reversible isomerization R ⇌ P in an isothermal, well-mixed chamber at fixed volume. Both species have the same molar mass, the total concentration c_T is constant, and the solution is ideal over the stated operating range. Let $x = c_P/c_T$, with $0 < x < 1$. Positive first-order rate constants k_f and k_r have units s^{-1}. The retained ideal activity convention is $a_R = 1-x$ and $a_P = x$, corresponding to an ideal binary mixture or the declared concentration reference $c^\circ = c_T$. The standard-potential difference and the kinetic equilibrium constant satisfy $\mu_P^\circ - \mu_R^\circ = -RT \ln(k_f/k_r)$. This ties the two μ quantities to the same reaction that supplies the chemical clock.

Equation 16: reaction law, time law and chemical potentials

$$dx/d\tau_a = k_f(1-x) - k_r x = \lambda(x_{eq} - x)$$
$$\lambda = k_f + k_r, \quad x_{eq} = k_f / \lambda$$
$$x(\tau_a) = x_{eq} + (x_0 - x_{eq}) \exp(-\lambda \tau_a)$$
$$\tau_c(x) = -(1/\lambda_{ref}) \ln[(x-x_{eq})/(x_0-x_{eq})]$$
$$\mu_R = \mu_R^\circ + RT \ln(1-x), \quad \mu_P = \mu_P^\circ + RT \ln(x)$$

[0068] On a calibrated monotone segment starting at $\tau_a = \tau_c = 0$, with $\lambda_{ref} = \lambda$ and the stated conditions, substituting the reaction solution into the chemical-time expression gives $\tau_c = \tau_a$. The equality follows algebraically from the physical rate law and its initial condition. The chamber sensor supplies x; the atomic channel supplies τ_a. Neither channel is populated by copying the other channel. In subsequent operation their measured residual tests whether the retained relation remains satisfied.

Equation 17: TAU TAU MU MU closure for R ⇌ P

$$r_\tau = \tau_c - \tau_a$$
$$j_+ = c_T k_f(1-x), \quad j_- = c_T k_r x$$
$$r_\mu = (\mu_R - \mu_P) - RT \ln(j_+/j_-)$$
$$r_x = x_{obs} - [x_{eq} + (x_0-x_{eq}) \exp(-\lambda \tau_a)]$$
ideal calibrated segment: $r_\tau = 0,\ r_\mu = 0,\ r_x = 0$

Equation 17A: the four-quantity physical differential binding

$$d\tau_c/d\tau_a = \{j_- / [c_T \lambda_{ref} (x_{eq}-x)]\} \times \{\exp[(\mu_R-\mu_P)/(RT)] - 1\}$$
for $0 < x < 1$, $x \neq x_{eq}$, $\lambda_{ref} > 0$ and fixed calibration
calibrated R ⇌ P model: $d\tau_c/d\tau_a = \lambda/\lambda_{ref} = 1$

[0069] Equation 17A directly binds both τ quantities and both μ quantities. It follows by differentiating the chemical-time map, substituting $dx/d\tau_a = (j_+-j_-)/c_T$, and applying the detailed-balance rate ratio. Its right-hand side is dimensionless. Together with the initial clock epoch, it yields the elapsed-time identity on the calibrated segment. The observability bound excludes the equilibrium point at which this inverse clock ceases to resolve time.

[0070] The logarithmic rate-ratio relation applies to the stated reversible reaction with compatible standard potentials and local detailed balance. More complex mechanisms retain a separate affinity and net reaction rate for each elementary reaction. A chemical species concentration, stress tensor, signed distance, chemical potential, and clock reading remain distinct field types throughout the lattice pipeline.

10.3 Physical identity, observation and acceptance

[0071] Each equation is accompanied by its operating domain and a residual definition. Algebraic identities and exact constants are evaluated as such. Observations carry measurement uncertainty, correlation and delay. The admission rule evaluates residuals against a retained uncertainty and tolerance budget; it also requires mass conservation, permitted temperature and pressure, identifiable chemical progress, and an accepted spatial registration. A failure retains the last admitted generation and records the failed condition. This makes physical agreement an executable criterion.

Equation 18: observation admission

$$\begin{aligned}\mathrm{admit}(O) = {} & \mathrm{unit_valid} \wedge \mathrm{calibration_valid} \wedge \mathrm{chronology_valid} \\ & \wedge |r_\tau| \le \varepsilon_\tau \wedge |r_x| \le \varepsilon_x \wedge |r_\mu| \le \varepsilon_\mu \\ & \wedge \mathrm{mass_valid} \wedge \mathrm{observable} \wedge \mathrm{operating_bounds_valid}\end{aligned}$$

[0072] A residual is tagged with its evidence dependencies. When both μ values and both rate terms are calculated from the same composition sample and model, r_μ verifies consistency of that calculation. A separately sensed or independently characterized energetic channel carries its own provenance. The record preserves this distinction so that correlated calculations are not counted as separate observations. The same rule applies to a fitted clock calibration and the samples used to fit it.

11. Atomic reference and chemical clock realization

11.1 Atomic timing channel

[0073] The SI second is defined using the fixed numerical value 9,192,631,770 Hz for the unperturbed ground-state hyperfine transition frequency of caesium 133 [T5]. An atomic-reference assembly provides a phase or frequency output, lock status, calibration record, and a time-transfer channel to the acquisition hardware. The transition defining the unit, the realized oscillator output, the signal divider, and the acquisition timestamp are represented separately.

Equation 19: atomic reference conversion

$$\begin{aligned}&\nu_{Cs} = 9\,192\,631\,770\ \mathrm{Hz} \\ &\Delta\tau_a = \Delta N_{ref} / \nu_{Cs} \\ &\Delta N_{ref} = \text{calibrated reference-cycle count including fractional phase} \\ &\text{observation epoch} = \text{trigger epoch} + \text{transfer-delay correction}\end{aligned}$$

[0074] The count conversion in Equation 19 uses the calibrated phase corresponding to the reference transition. For an apparatus exposing a divided output such as a frequency signal and a pulse-per-second signal, its ratio and phase relationship to the reference are part of the adapter. Phase, frequency and oscillator comparison have the physical relationship described by NIST [T9]. The adapter stores the raw count, fractional phase, realized frequency correction, lock history, uncertainty, signal path,

trigger convention, and counter rollover behavior. Integer counters remain integers through subtraction; a large epoch is not first rounded into a floating-point number.

[0075] Clock comparisons are made at a declared reference location and epoch. Separate clock locations add the time-transfer and relativistic rate corrections appropriate to the installation. Reaction-duration control uses elapsed intervals, so the apparatus also retains a monotonic counter even when a civil-time label is attached. An acquisition timestamp identifies the sample aperture or a defined point within it, and the integration duration is recorded.

11.2 Chemical timing channel

[0076] A chamber sensor measures chemical composition through a calibrated observation function y = H(c,T,p) + noise. A spectroscopic, electrochemical or other species-selective transducer may be used. The adapter converts its raw voltage, intensity or spectrum to activities or concentrations with the calibration identity retained. For the R ⇌ P embodiment, a composition estimate x determines τ_c through Equation 16. Calibration estimates k_f, k_r, x_0 and the equilibrium composition under controlled conditions and retains parameter covariance and calibration samples.

[0077] The inverse chemical clock requires $x_0 \neq x_{eq}$, $\lambda_{ref} > 0$ and $0 < (x-x_{eq})/(x_0-x_{eq}) \leq 1$ for a forward elapsed-time segment. The measured x remains on the declared side of x_eq and within the calibrated monotone segment. A finite-resolution sensor loses time sensitivity near equilibrium. The implementation therefore carries a minimum separation from equilibrium and a maximum propagated time uncertainty. It renews the segment before either bound is crossed. Temperature changes either invalidate the segment or invoke an independently retained temperature-dependent kinetic model; the clock is not silently renormalized to its atomic comparator.

Equation 20: local chemical-time sensitivity

$$|\partial \tau_c/\partial x| = 1 / [\lambda_{ref} |x - x_{eq}|]$$
$$u^2(\tau_c) = J_c \Sigma_{cal,obs} J_c^{\top}$$
$$u^2(r_\tau) = u^2(\tau_c) + u^2(\tau_a) - 2\,\mathrm{cov}(\tau_c,\tau_a)$$
$$\text{observable} \Leftrightarrow |x-x_{eq}| \geq \delta_{eq} \text{ and } u(\tau_c) \leq u_{max}$$

[0078] J_c is the gradient of the chemical-time expression with respect to the observed composition and retained calibration parameters; Σ_cal,obs includes their covariance. The composition-only derivative gives a useful sensitivity check, while the full propagation includes the fitted rate and equilibrium uncertainty. The residual budget also includes timestamp aperture, sensor latency, transport lag, and spatial interpolation when they affect the comparison.

11.3 Reset, recurrence and oscillator embodiment

[0079] A single R ⇌ P relaxation supplies one monotone chemical-clock segment. A physical reset flushes, recharges or otherwise restores a specified initial composition, measures that composition, and establishes a new segment epoch. The global chemical clock is $\tau_{c,global} = T_{epoch} + \tau_{c,segment}$, with T_{epoch} tied to the measured handover event. Reset duration and unobserved intervals remain in the epoch record. The feed amounts, waste, heat and work consumed by resetting are entered in the physical balance.

[0080] An oscillatory embodiment replaces the monotone inverse by a phase reconstruction from a measured chemical limit cycle. Each unwrapped cycle has a cycle index and an identified reference crossing; a hysteresis band and refractory interval prevent duplicate crossings. Chemical phase time is computed from that phase and a retained period model. A driven chemical oscillator records its fuel and boundary conditions. Cao and co-authors analyze measured KaiC phosphorylation oscillations under different ATP/ADP conditions as a biochemical timing example [T8]. The architecture uses the measured waveform and its calibration without assuming that all chemical reactions oscillate.

Equation 21: a calibrated phase-clock embodiment

$$\theta_c = 2\pi\, m + \theta_{within_cycle}, \quad m \in \mathbb{Z}$$

$$\tau_c = (\theta_c - \theta_{c,0}) / \omega_{ref}$$

$$r_{phase} = \mathrm{wrap}(\theta_c - \omega_{ref}\, \tau_a - \theta_{target})$$

missing or ambiguous crossings invalidate the affected interval

12. Coupled physical balances and the spatial lattice

12.1 Material, energy and mechanical equations

[0081] A physical lattice can contain interconnected reaction chambers, transport channels and load-bearing material. In a spatial embodiment, a reaction-transport model updates species concentrations in the retained physical frame. Species source terms, imposed fluxes and moving boundaries are explicit. A reaction stoichiometric matrix is mass balanced before its rates are used. The chemical clock at a point is derived from the local reaction state or from a declared mixing-volume observation.

Equation 22: species balance and a declared flux law

$$\partial c_i/\partial \tau_a + \nabla\cdot N_i = \Sigma_r\, \nu_{ir}\, j_r + s_i$$

$$N_i = c_i\, v - [D_i\, c_i/(RT)]\, \nabla\mu_i$$

$M^T\nu = 0$ for closed mass-balanced reactions

$[c_i] = \mathrm{mol\ m^{-3}}$; $[N_i] = \mathrm{mol\ m^{-2}\ s^{-1}}$; $[j_r] = \mathrm{mol\ m^{-3}\ s^{-1}}$

[0082] The displayed flux expression is an isothermal scalar-mobility constitutive embodiment. Other transport laws retain their own assumptions and coefficients. At fixed temperature and pressure, a closed reaction with extent ξ in moles satisfies

dG/dτ_a = -A_r dξ/dτ_a. At fixed temperature and volume, the corresponding identity uses Helmholtz free energy F_H: dF_H/dτ_a = -A_r dξ/dτ_a. The reaction entropy-production contribution is A_r(dξ/dτ_a)/T. For the detailed-balanced elementary reaction in Section 10 this contribution is nonnegative. A physical reset receives free energy through supplied matter or external work; its balance covers the reservoirs as well as the clock chamber. Primary reaction-network thermodynamics also treats chemical work and entropy accounting [T10]; this embodiment uses deterministic molar quantities.

Equation 23: reaction and control-volume energy accounting

fixed T,p: dG/dτ_a = -A_r ξ; fixed T,V: dF_H/dτ_a = -A_r ξ
S_i,reaction = A_r ξ/T ≥ 0
dU/dτ_a = Q_in + Ẇ_in + Σ_in ṅᵢ ħᵢ - Σ_out ṅᵢ ħᵢ
r_E = ΔU - integral of the recorded net energy input

[0083] Q_in and Ẇ_in are heat and non-flow work rates into the control volume under the stated sign convention; flow work is included in the molar enthalpy terms. The displayed internal-energy balance takes kinetic and gravitational energy changes as negligible; a moving or vertically extended apparatus includes those terms in its total-energy balance. Chemical reaction changes are included in the material internal energy and species enthalpies, so affinity is not added a second time as an independent energy input. Mechanical loading is coupled through a declared constitutive law and the force balance ∇·σ + b = ρ ü. Its boundary conditions and material parameters identify the biomechanical operating window.

12.2 Physical state to signed distance

[0084] The chemical channel forms an operating set from physical predicates. An embodiment requires a specified activity interval, a minimum reaction affinity, and an admitted timing residual. Its boundary is then represented by φ_A = SD(Ω_A). Stress or strain predicates form Ω_B and φ_B as in Sections 2 and 3. The membership-to-distance conversion establishes length-valued geometry from a physical domain predicate; it does not relabel a chemical potential or a concentration as a distance.

Equation 24: bound physical windows and retained lattice construction

Ω_A(e) = {x : a_min ≤ a(x,e) ≤ a_max, A_r(x,e) ≥ A_min,
|r_τ(x,e)| ≤ ε_τ and observation_valid(x,e)}
φ_A = SD(Ω_A), φ_B = SD(Ω_B)
A = w_A(φ_A), B = w_B(φ_B), S = (A ⊙ B) ⊙ C
Ω_V = Reg((Ω_L ∩ {S ≥ τ_g}) \ Ω_P),
Ω_F = Reg(Ω₀ \ Ω_V)

[0085] A distinct control embodiment keeps the spatial windows fixed and uses observed chemical progress to choose τ_g, a lattice radius, a sheet thickness or a bounded actuation dose from the LUT. The mapping is an explicit control design with physical input units and output units. For example, a progression table can select increasingly permissive score thresholds as a timed reaction advances. Reaction kinetics determines

the observed progress; the controller policy determines which geometry is requested at that progress.

12.3 Closing the physical geometry loop

[0086] A build or removal actuator receives a geometry-linked command in a registered physical frame. The command records feedstock or removed material, spatial bounds, duration, rate limits and the accepted clock interval. A receipt records actual execution identifiers and times. A subsequent shape or mass measurement establishes the realized result. Proposed geometry, actuator receipt and observed geometry have separate identities. A mesh calculation alone produces the proposed-geometry identity.

Equation 25: structural realization residuals

$$r_m = \rho_s (V_{after} - V_{before}) - \int (\dot{m}_{deposit} - \dot{m}_{remove})\, d\tau_a$$

$$r_G = d_H(\partial\Omega_{commanded}, \partial\Omega_{observed})$$

realization accepted when material, shape and timing budgets hold

[0087] The mass expression uses the same constant density ρ_s before and after the operation. Variable-density material uses the difference of the after-state and before-state density volume integrals. d_H denotes a specified Hausdorff boundary-distance comparison on registered surfaces, supplemented by topology and protected-region checks. Geometry, material flow and clock records therefore constrain the same physical transformation. Measured transport or mechanical response can subsequently renew the chemical and biomechanical windows, completing the cross-domain feedback path.

13. Autopoietic self-referential lookup tables

13.1 Organization, entries and self-description

[0088] The lookup table L_n is the working organization of generation n. It contains typed physical-state rows, bounded control rows, prediction rows, and meta-rows describing its own schema, units, dependency graph and regeneration recipe. A lookup key is produced by a declared quantizer over selected components of B_{phys} and associated state. Quantization intervals, inclusive boundaries, interpolation and out-of-range behavior are part of the table definition. Unquantized observation values and their uncertainty are retained alongside their quantized keys.

Row family	Required contents	Operational role
Physical binding	τ_a, τ_c, μ_R, μ_P; location; epoch; raw-observation and calibration references.	Connect the entry to a measured physical state.
Prediction	Future composition or phase; admissible residual interval; kinetic model and horizon.	Generate a subsequently testable physical expectation.

Row family	Required contents	Operational role
Control	Input interval; τ_g or geometric parameter; permitted action; resource and output units.	Map admitted physical state to a bounded command.
Schema / unit meta-row	Column names, field types, quantity dimensions, version and decoding rules.	Describe and regenerate the working table structure.
Regeneration meta-row	Recipe identifier, seed inputs, dependency digests, parent link and update bounds.	Reproduce the table and generate its successor.
Self-reference	A symbolic reference to the current verified body and a link to its predecessor.	Resolve the table's own identity without recursive hashing.

[0089] The stable interpreter K defines the meanings of field types, physical units, permitted actions, digest verification and commit ordering. The LUT supplies the regenerable organization interpreted by K. A table may change its predictive rows and control entries under its retained recipe; a separately versioned migration is required to change the interpretation of its schema. This finite execution contract terminates the self-reference and makes reconstruction executable.

Equation 26: autopoietic regeneration

recipe_n = resolve(L_n.regeneration_recipe, K)
L_candidate = recipe_n(L_n, O_post, G_observed, resources)
organization(L_candidate) satisfies the retained invariant contract
L_(n+1) = commit(L_candidate) when physical and structural checks pass

13.2 Productive closure

[0090] Regeneration performs work: it reconstructs the schema rows from the retained channel registry, regenerates the applicable physical forecast rows, rebuilds control intervals and dependency references, links the selected geometry, and emits the next self-describing body. The generated rows can be deleted from a working copy and reconstructed from the retained recipe, complete source observations and seed state. Repeating the regeneration with identical inputs must reproduce the same canonical body. New observations may produce a different body, while the organizing rule and invariant boundary persist.

[0091] The physical realization adds a resource-renewal branch. A maintenance or recharge command restores a measured chemical initial condition, replaces a consumable component, renews a sensor calibration, or restores an allowed structural condition. The replacement becomes part of the next organization only after its measured state satisfies the same physical binding contract. The scope of self-production is specified by the component registry; each component states whether it is regenerated as a table record, recharged chemically, fabricated materially, or maintained by an external service.

13.3 Self-reference and reproducible identity

[0092] A self-reference is represented as a symbolic current-body reference resolved by a verified envelope. The hashed body includes the parent digest, generation, recipe, rows, observations and dependency identities. Its own digest field is outside the hashed body. This gives a finite construction: serialize the body, compute its digest, and place both in the envelope. The next generation contains the preceding digest as an ordinary parent value. A flat envelope may place body members and its reserved digest at the same top level; the hashed body is then the mapping obtained by omitting that digest member. The reference implementation uses this flat encoding and an executable current-body resolver.

Equation 27: finite self-reference

```
body_n = {generation, parent_digest, recipe, rows, dependencies,
          self_ref: CURRENT_BODY}
h_n = SHA256(canonical_encode(body_n))
envelope_n = {body: body_n, body_digest: h_n}
resolve(CURRENT_BODY, envelope_n) = h_n
```

[0093] Canonical encoding fixes key ordering, character encoding, separators, finite-number representation and container ordering. The reference implementation documents its actual encoding profile. Full snapshot bodies and recipe inputs are retained because a digest identifies bytes but does not contain the bytes needed to reconstruct a lost table. An implementation using another serialization standard must preserve the same declared identity contract across all participants.

13.4 Admissible adaptation and recovery

[0094] A candidate cannot change the physical meaning of τ_a, τ_c, μ_R or μ_P, broaden its own acceptance threshold to pass a failed observation, replace an observed quantity with a prediction, or point its parent to a different active generation. A model update has an explicit data interval and a subsequent validation interval. Parameter changes remain within declared rate, temperature, geometry and actuator bounds. The previous admitted state and failed attempt are retained for recovery.

[0095] Recovery verifies the saved body digest, resolves its complete dependencies, and invokes a same-generation rebuild with the saved generation number, parent and seed. It compares the rebuilt body and linked geometry records with the retained generation. Advancing the current state to a successor generation is a separate operation with a new generation number and parent link. A missing generated row is repaired from the retained organization; a missing raw observation is identified as a missing dependency. A table cannot manufacture evidence for an absent physical observation. A successful reconstruction returns an executable generation with the same meaning and identity as the saved generation.

14. Physical control, regeneration and operating cycles

14.1 Observation-to-action closure

[0096] The apparatus repeatedly acquires a physical observation, evaluates the four-quantity binding, constructs a candidate target and obtains its physical realization. This control lifecycle uses ordinary transaction stages. TAU TAU MU MU retains its four-quantity meaning throughout every stage. The active table is the latest generation whose required observations and checks have completed.

Equation 28: physical operational closure

u_n = policy(L_n, O_pre, G_current)
z_post = PhysicalEvolution(z_pre, u_n, Δτ_a, resources)
O_post = Observe(z_post; sensor_calibrations)
L_candidate = Regenerate(L_n, O_post, G_observed)
L_(n+1) = Commit(L_candidate; receipt, residuals, invariants)

Stage	Concrete output	Completion criterion
Acquire and bind	Raw observations, B_phys, residual vector and uncertainty record.	Correct epoch, units, calibrations and operating domain.
Propose and validate	Candidate geometry, control values, transaction ID and expected post-state.	Physical model, resources, protected regions and actuator bounds satisfied.
Apply and receive	Command linked to active LUT and target geometry; execution receipt.	The addressed device reports its actual execution interval and result code.
Observe result	New chemical, geometric or mechanical measurements.	Required observables are acquired after the relevant physical action.
Regenerate and commit	Successor table, observation links, receipt link and parent digest.	Post-state satisfies the recorded physical and structural contract.

[0097] A receipt establishes which command an actuator accepted or executed. The subsequent observation establishes the physical result required by that command. If only chemical evolution is exercised, the receipt and post-observation describe that chamber interval and the geometry remains a linked design output. A fabrication realization additionally supplies geometric metrology and material-flow evidence. The transaction records its exercised mode.

14.2 Bounded adaptation of the physical plant

[0098] The controller may adjust temperature, reactant feed, flow, catalyst availability, illumination, mechanical boundary loading or fabrication parameters when that actuator is present and its constitutive model is identified. Each action has units, admissible bounds, timing, resource accounting and a measured completion condition. If an action changes the reaction rate or equilibrium state, its prediction uses the updated dynamic model over the actual actuation interval; the constant-rate inverse of Equation 16 is used only on its declared segments.

[0099] For a chemical oscillator, an identified input can regulate its phase relative to the atomic reference. A controller may use a calibrated phase-response model and a bounded input proportional to the wrapped phase residual. The transfer function, actuation delay, saturation, stability range and source of chemical free energy belong to that embodiment. Phase locking is established by the retained phase residual and cycle record.

14.3 Physical regeneration and resource closure

[0100] A regeneration command identifies the component being renewed, the consumed resource, the expected restored state, and the observations needed to resume operation. For a relaxation clock this may be a chamber recharge. For a lattice it may be replacing a sacrificial insert, controlled deposition, removal of a clogged region, or changing a reversible boundary condition. Regeneration creates a successor component record linked to both the old component and the actual material or chemical operation.

[0101] The epoch policy states whether global chemical time counts only active chemical segments or also contains explicitly labeled atomic bridging intervals during recharge. Both policies retain the recharge start and end; the same interval is not silently inserted into a chemically measured duration. Incoming initial composition and outgoing final composition are measured. The table regenerates the next clock segment from the observed restored state.

14.4 Interrupted actions and persistent physical state

[0102] A transaction identifier makes command retries idempotent at the addressed device. After an interrupted connection, the controller checks that identifier and acquires the current physical state before deciding whether another action is required. A failed validation can discard an unexecuted proposal. A failed post-observation leaves an executed action in the physical ledger even when the candidate table is not committed. The next recovery proposal starts from the observed plant state, since material removal, reaction progress and elapsed time do not roll back with a table pointer.

15. Records, calibration and executable contracts

15.1 Required physical records

Record	Required fields
AtomicReferenceRecord	Reference species/transition or traceable source; output ratio; raw integer count and fractional phase; delay correction; lock status; units; calibration; uncertainty.
ReactionModelRecord	Species and balanced stoichiometry; standard states; activity model; rate law; coefficients and covariance; temperature/pressure range; initial conditions; material and energy ports.
ChemicalObservation	Raw sensor output; calibration mapping; composition or phase; acquisition aperture; latency; location; units; uncertainty; source mode and sensor identity.
PhysicalBindingRecord	Both τ quantities, both μ quantities, reaction affinity, calibration versions, source dependencies, residuals and admission result.
LUTGeneration	Canonical body; schema and recipe; predictive/control/meta-rows; parent and current digests; complete reconstruction dependencies; active generation.
ActuationRecord	Transaction and device identities; command and geometry digests; bounded setpoints; resource plan; actual receipt; command/receipt times.
PostObservationRecord	Post-action chemical and physical measurements; measured geometry when required; closure residuals; receipt and pre-observation links.
RegenerationRecord	Component scope; restored initial condition; supplied/removed matter and energy; clock-segment epochs; successor component/calibration references.

[0103] A field produced by an instrument, an imported record, an analytic model or a numerical solver has an explicit source mode. Model-derived fields retain their input dependencies. Counters larger than the exact integer range of a consumer are transported as decimal integer strings or a lossless integer encoding. Numeric NaN and infinity, unknown units and unresolved calibration identifiers are invalid values for an admitted binding.

15.2 Deterministic execution contract

```
active = verify_and_restore(saved_generation)
pre = acquire_observation(atomic_adapter, chemical_adapter)
binding = bind_physical_quantities(pre, active.calibration)
require admissible(binding, active.fixed_acceptance_policy)
proposal = make_bounded_proposal(active, binding, geometry)
require model_and_geometry_checks(proposal)
receipt = actuator.apply_once(proposal.transaction_id, proposal)
post = acquire_required_post_observation(receipt)
candidate = regenerate(active, post, receipt, proposal)
require physical_closure(candidate, post, receipt)
require candidate.parent_digest == active.body_digest
persist_complete(candidate); active = commit(candidate)
```

[0104] The regeneration rule is a deterministic function of its retained inputs. Physical evolution and observation occur outside that deterministic reconstruction. Replay therefore consumes the saved observation and receipt; it does not generate replacement observations. The numerical package includes an analytic-generated plant adapter and a recorded-observation input format so that the execution boundary is visible and reproducible.

15.3 Acceptance measurements and retained scope

[0105] Clock acceptance uses an independently obtained composition or phase observation against a retained calibration. Kinetic acceptance compares observed composition change with a predicted change over an interval. Thermodynamic checks compare potential/rate consistency and the sign of the reaction entropy-production contribution. Material checks compare species inventories and net flux. Geometry checks compare the registered proposed and observed boundaries when physical fabrication is exercised. Each check records whether it is an algebraic consistency test, a model prediction test, a sensor comparison or a material realization measurement.

[0106] The invariant contract separates numeric equality used to reproduce a canonical table from engineering tolerances used to compare observations. Bitwise replay addresses table identity. Residuals and uncertainty address physical agreement. Dimensional validation addresses the meaning of a field. All three are necessary for the same observation-to-lattice operation to remain interpretable after regeneration.

16. Worked clock-bound lattice construction

16.1 Declared numerical apparatus

[0107] The executable construction uses an analytic-generated R ⇌ P plant and a generated atomic-reference count channel. The chemical sample is generated from the physical reaction equation, then passed through the same observation, inverse-clock, affinity, residual and LUT interfaces used for recorded input. Source fields identify the generated channels and their shared dependencies. Integer tick conversion and floating-point reaction evaluation are recorded separately.

Quantity	Specified value or relation
Temperature	298.15 K
Forward / reverse rate constants	$k_f = 0.03\ s^{-1}$; $k_r = 0.01\ s^{-1}$
Relaxation rate / equilibrium fraction	$\lambda = 0.04\ s^{-1}$; $x_{eq} = 0.75$
Initial product fraction	$x_0 = 0.10$
Standard-potential reference	$\mu_R^\circ = 0$; $\mu_P^\circ = -RT \ln(3)$, in J mol^{-1}
Gas constant used	$R = 8.31446261815324$ J mol^{-1} K^{-1}
Atomic reference count conversion	$\nu_{Cs} = 9{,}192{,}631{,}770$ cycles per SI second
SDF structural base	12 × 12 × 8 mm; 0.5 mm cells; orthogonal channel network
Geometry-control values	Dimensionless τ_g values: 0.75, 0.65, 0.55, 0.45, 0.35
Input / receipt source	Analytic-generated observations and generated plant receipts; recorded-input adapter included

Equation 29: explicit physical binding of the worked construction

$$x(\tau_a) = 0.75 - 0.65 \exp(-0.04\ \tau_a)$$
$$\tau_c = -25 \ln[(0.75 - x)/0.65] \text{ seconds}$$
$$\mu_R = RT \ln(1-x), \quad \mu_P = RT \ln(x/3)$$
$$\mu_R - \mu_P = RT \ln[3(1-x)/x]$$
$$\tau_c[x(\tau_a)] = \tau_a \text{ on the retained clock segment}$$

[0108] For example, after 25 seconds the remaining displacement from equilibrium is 0.65/e, so the analytic-model fraction is x = 0.75 - 0.65/e and the inverse returns 25 seconds. The affinity at that state is obtained from the same x through the displayed potential difference. This makes the two time quantities and two chemical potentials a single physically specified relation, with their distinct units and meanings intact.

16.2 Regenerating the table and linked geometry

[0109] The generated observations advance the table through admitted physical states. Each regenerated body contains its schema, physical binding, predictive entries, control entries, parent link and reconstruction dependencies. Its selected τ_g is applied to the original window-gate SDF lattice with the same unclipped-cutter convention. The resulting geometry record is linked to that generation. The complete control interval definitions are stored in the executable policy and exported table.

[0110] The time-driven geometry selection is a specified controller policy over measured chemical progress. Its values choose different subsets of the same channel lattice. The reference chamber supplies the timing state; the example retains the spatial window fields from Section 9. This isolates and exercises the timing-to-LUT-to-geometry

connection while the apparatus embodiments in Sections 12, 14 and 17 specify measured spatial feedback and physical structural realization.

Integrated execution result	Recorded value
Integrated verification suite	24 tests passed
Admitted table generations	6 including the initial generation
Completed successor commits	5
Exercised fault responses	3 holds; 2 rejections
Distinct generated lattice geometries	5
Regenerated predictive row counts	12 → 12 → 10 → 8 → 7 → 6
Maximum absolute clock residual	2.35438e-11 s
Maximum |dτ_c/dτ_a - 1|	8.88178e-16
Maximum composition residual	6.09068e-13
Full-history reconstruction	Identical verified head
Generated geometry topology	All reported closed oriented manifold checks passed
Retained volume, first / final	962.5 / 820.5 mm³
Evidence source	Analytic-generated reaction; generated count and actuator channels

16.3 Reproducible verification

[0111] The package records the executed test results and numerical output. The physical-clock checks cover analytic inversion, reaction affinity and rate consistency, admissible operating range, uncertainty growth near equilibrium and rejection of inconsistent observations. The regenerative checks cover complete reconstruction, parent identity, tamper rejection and the requirement for a linked receipt and post-observation. The geometry checks use actual sampled masks and their volume partition.

[0112] The earlier SDF and CSG constructions remain in the package with their original execution records. The new integrated run identifies the stages it executes, the channel source modes and the outputs it generates. A user can replay the retained observations and regenerate the same table bodies, residuals and selected geometry without inventing missing measurements.

17. Apparatus embodiments and construction details

17.1 Reference-compartment apparatus

[0113] A principal apparatus uses a fixed-volume reference reaction chamber alongside a lattice specimen or fabrication region. The chamber includes temperature measurement and control, a composition sensor, a calibrated acquisition aperture, and supply/waste ports for renewal. The atomic reference timestamps chamber observations and actuator events. The specimen has its own registered chemical, mechanical and geometric observations. Their relationship to the reference chamber is recorded as a sampling, shared-medium, transport or control connection.

[0114] Keeping the timing chamber at its stated volume and boundary conditions preserves the analytic clock calibration while the specimen geometry changes. If the clock reaction instead occurs within a changing lattice volume, the model includes dilution, transport, moving boundaries and their measured control inputs. The fixed-volume expression is then replaced by the corresponding physical observation model. A local chemical clock is a field attached to location; spatial differential operators continue to use the declared physical time coordinate.

Assembly	Physical connection and specified interface
Atomic reference and counter	Frequency/phase source; divider calibration; lossless counter; lock and time-transfer status.
Reference reaction compartment	Defined volume and initial composition; T and p; species-selective observation; recharge ports.
Spatial chemical sensing	Activity/concentration/phase observation registered to the specimen with sampling and transport delay.
Mechanical and geometric sensing	Load, displacement or strain; shape/mass metrology; common spatial calibration.
Regenerative LUT controller	Typed physical binding, retained schema/recipe, prediction, bounded policy, full snapshots and event ledger.
Geometry or boundary actuator	Material deposition/removal, fabrication, reversible deformation or boundary-control interface; command receipt.
Material and energy ports	Metered feed/waste; electrical/mechanical work; heat exchange; component-renewal records.

17.2 Shared free-energy and moving-boundary embodiments

[0115] A small-strain isothermal chemoelastic embodiment derives species potentials and stress from one retained Helmholtz free-energy density $\psi(c,\varepsilon,T)$ per reference volume. With ψ measured in J m^{-3} of reference volume, concentration in mol m^{-3} of the same reference volume and strain dimensionless, the derivatives give μ_i in J mol^{-1} and the small-strain stress σ in Pa. For each locally detailed-balanced elementary reaction step, reaction flux ratios use the resulting total affinity. This prevents a mechanical contribution to chemical driving from being added to the potential while leaving an incompatible kinetic equilibrium relation unchanged.

Equation 30: shared energetic and boundary evolution embodiment

$\mu_i = \partial\psi/\partial c_i, \quad \sigma = \partial\psi/\partial\varepsilon$

$j_+/j_- = \exp[(\mu_R - \mu_P)/(RT)]$

$\partial\varphi/\partial\tau_a + v_n\,|\nabla\varphi| = 0$

v_n is measured or derived from a recorded material-transfer law

[0116] The boundary equation uses a sign and normal convention retained with the level set. Positive v_n advances the boundary in its outward normal direction for the inside-negative convention. A deposition/removal law ties that motion to material transfer and density; a deformation law ties it to displacement. Reinitialization restores distance values when required without changing the intended boundary. The lattice and its physical transport and mechanical coefficients are recalculated after an admitted boundary change.

17.3 Distributed, sampled and continuously operating embodiments

[0117] A distributed implementation maintains one authoritative parent chain per controlled physical region and records synchronization between regions. Atomic time transfer supplies interval correspondence; chemical clocks retain their local calibration, cycle or segment number. Message delay and stale observations are checked before actuation. A shared geometric boundary has one declared owner or an explicit coordination rule so that two regenerative tables cannot issue incompatible material commands for the same interval.

[0118] Sampled LUT entries may be evaluated by direct interval lookup or by interpolation within a recorded valid cell. Interpolation preserves units and bounded control outputs. It does not convert a nonmetric implicit field into a metric SDF. A continuous-state implementation may evaluate the same regeneration recipe on demand and cache the resulting entries with identical physical and self-reference records.

[0119] The integrated architecture therefore supports chemically timed fabrication, regenerative microfluidic lattices, monitored scaffold geometry, reversible structural control, and cyclic chemical-clock renewal. Each embodiment instantiates the relevant material, sensing and actuation interfaces and retains the common quartet, spatial registration, table-regeneration rule and physical closure contract.

18. Description of drawings and reference numerals

[0120] FIG. 1 shows the computing architecture. Two domain windows enter the common registration and SDF layer. Gate mapping and the double Hadamard coupling control the lattice and negative-solid stages. The ordered CSG evaluator combines the base and negative solid, and an optional return path supports geometry-driven field updates.

[0121] FIG. 2 shows the typed field algebra. Signed-distance operands are mapped to dimensionless gates before the two Hadamard operations. A separate geometric branch supplies the lattice field. The selected score region and lattice field combine into the negative solid, which is subtracted from the base.

[0122] FIG. 3 shows the execution flow and its branch conditions. Input validation precedes coupling. Membership reconstruction resolves threshold plateaus and boundary cases. The result passes through geometry checks before output or an iteration update.

[0123] FIG. 4 shows the base, the field-selected negative lattice, and the retained solid of the SDF numerical construction. The panels use the same physical scale and orientation. The negative solid occupies the channel region selected jointly by the domain windows and lattice generator.

[0124] FIG. 5 shows a spatial slice through the two window gates, structural modulation, coupled score, selected lattice cells, and retained material. The common grid makes the local relationship among the operands and output directly visible.

Numeral	Element
102 / 104	Biochemical / biomechanical domain input
110	Spatial registration and typed field records
120 / 130	Signed-distance windows / gate maps
140 / 142 / 144	Coupling layer / first product / second product
150	Lattice generator
160	Negative-solid constructor
170 / 180	CSG difference evaluator / geometry output
190	Geometry-driven update path
302 / 304	Input acquisition / registration check
308 / 310 / 312	Gate mapping / first product / second product
314 / 316	Membership and lattice selection / solid reconstruction

Numeral	Element
318 / 320 / 322	Difference evaluation / geometry checks / export
402 / 404 / 406	Base solid / negative lattice / retained solid

[0125] FIG. 6 sets out the physical TAU TAU MU MU binding, placing atomic time, chemical time, reactant potential and product potential around the reaction law and its closure residuals. FIG. 7 shows the reference clock, chemical chamber, regenerative controller, actuator and post-observation loop, including resource inputs. FIG. 8 shows finite self-reference and regeneration through retained schema, recipe, observations, canonical bodies and parent digests.

[0126] FIG. 9 plots the generated reaction-clock trajectory, chemical-time reconstruction and two chemical potentials from the integrated numerical construction. FIG. 10 relates the admitted physical progression to the selected geometry-control values and the resulting lattice volumes. These sheets identify generated data as part of their captions and retain the input units.

Reference	Element
502 / 504	Atomic reference assembly / lossless acquisition counter
510 / 512	Reference reaction compartment / composition or phase sensor
520	Physical TAU TAU MU MU binding and residual evaluator
530 / 532	Regenerative LUT body / schema and recipe meta-rows
540 / 542	Bounded command and geometry target / actuator receipt
550	Post-action physical observation and realization checks
560	Matter, energy and component-renewal interfaces
570 / 572	Verified generation envelope / predecessor reference

[0127] FIG. 11 shows inverse-clock conditioning, rate-scale mismatch, finite-population fluctuations and observation-aperture bias for the declared reference model. FIG. 12 shows the grid-resolution dependence of the SDF construction and a circular-channel benchmark. Both sheets distinguish generated sensitivity results from physical measurement.

19. Scope of physical exactness and evidence

19.1 Exactness has a declared object

[0128] An algebraic equality, a numerical result, a deterministic model and a physically accurate prediction are different objects. Exactness is stated with the object, assumptions, environment and evidence that support it. This edition uses the term exact model identity for a symbolic result under stated assumptions; numerical verification for implementation and discretization checks; and physical validation for comparison with independent measurements of the represented apparatus [T16].

Evidence class	Meaning	What it can establish
D: definition	Chosen units, conventions and exact defined constants.	Meaning and dimensional consistency.
A: analytic identity	A relation derived within an explicit model.	Logical consistency conditional on that model.
N: numerical verification	Code, solver, mesh and reconstruction checks.	Correct execution to a declared numerical tolerance.
M: calibrated measurement	Instrument observations with traceability and uncertainty.	An estimate of a physical measurand.
V: independent validation	Predictions compared with measurements outside model fitting.	Accuracy over the measured operating domain.

19.2 A complete mathematical instance

Equation 31: closed instance and coupled evolution

$$I_{phys} = (\text{domains, state } z_0, \text{ parameters } \theta, \text{ constitutive laws,}$$
$$\text{boundary data, inputs } u(t), \text{ sensors, timebase, tolerances})$$
$$\dot{z} = F(z,u,t;\theta) + \delta_{model}; \quad y_k = H_k[z,\theta] + \varepsilon_k$$
$$L_{(n+1)} = R_K(L_n, \text{retained observations, receipts, recipe})$$

[0129] The state includes reference-chamber composition, specimen species fields, temperature, displacement/velocity, boundary geometry, actuator state, reservoir inventories and clock/observer states when those channels are exercised. A spatial discretization produces an ODE or differential-algebraic system only after constraints, initial conditions and boundary conditions are supplied. δ_{model} denotes retained model discrepancy; omitting it from a reference calculation does not prove it is zero in an apparatus.

[0130] The reference implementation supplies a closed ideal reaction instance and a deterministic LUT/geometry pipeline. This extension supplies the constitutive and observation interfaces for physical instances. An unmeasured coefficient is identified as a required empirical input; an illustrative value is labelled as such and is not presented as a calibration.

19.3 The quartet is a relation, not four independent observations

[0131] For the ideal example, τ_c, μ_R and μ_P are all functions of the same composition x and retained parameters. They therefore do not provide three independent observations of the reaction. The quartet also requires a temperature, stoichiometry, kinetic scale and calibration. In particular, multiplying k_f and k_r by the same positive factor leaves equilibrium and the ideal potential difference unchanged while changing the reaction timescale. Chemical affinity alone does not determine an absolute clock rate.

Equation 32: kinetic-scale identifiability

(k_f,k_r) → (a k_f,a k_r), a > 0
x_eq and k_f/k_r unchanged; λ → aλ
μ_R - μ_P unchanged at the same x,T; relaxation time → 1/(aλ)

20. Timebase, sensing and observable quantities

20.1 Physical time and an observed clock

[0132] Let t denote the chosen local laboratory time coordinate for the plant equations. The measured atomic-reference elapsed-time estimate is written $\hat{\tau}_a$. The ideal identification t = τ_a used in Equations 16-17A is the zero-error reference case. Actual timestamps retain residual phase/frequency error, epoch error and time-transfer effects. The SI defining frequency is exact by definition; an instrument realization and transfer have uncertainty [T5, T13].

Equation 33: atomic observation and residual time error

$N_a(t) - N_a(0) = \int_0^t \nu_{out}(s)\,ds + \Delta\eta_{phase}(t)$
$\hat{\tau}_a(t) - t = b_a + \int_0^t y_a(s)\,ds + \eta_a(t)$
$\nu_{out,nom} = r_{out}\,\nu_{Cs}$; r_{out} is the retained output ratio
$u^2(\hat{\tau}_a)$ includes frequency, phase, count, epoch and transfer terms

[0133] The residual fractional-frequency error y_a is dimensionless; b_a and η_a have units s. A constant residual y_a creates time error y_a t. A counter increment is divided by its actual calibrated output frequency, not automatically by ν_Cs. Location and time-transfer corrections are applied before combining channels; they are not absorbed by fitting the chemical clock to the same validation data.

Equation 34: weak-field clock comparison when required

$d\tau_{proper}/dt_{coord} \approx 1 + \Phi_g/c^2 - |v|^2/(2c^2)$
relative rate A-B $\approx (\Phi_{g,A}-\Phi_{g,B})/c^2 - (v_A^2-v_B^2)/(2c^2)$

[0134] Equation 34 uses a weak, slowly varying field, gravitational potential Φ_g with the usual negative-potential convention, a specified coordinate time, and corrections through order c^{-2}. It is a relative comparison rule for separated/moving clock systems.

A local proper second is not corrected merely because the laboratory sits in a gravitational field. An installation selects the applicable time-transfer model [T13].

20.2 A sensor measures an aperture, not an instantaneous state

Equation 35: sampled observation operator

$$y_k = (1/\Delta_k) \int_{[t_k - d_k - \Delta_k/2,\ t_k - d_k + \Delta_k/2]} h(z(s);\theta_h)\, ds + b_k + \varepsilon_k$$

$\Delta_k > 0$: aperture; d_k: latency; b_k: bias; $\mathrm{cov}(\varepsilon) = \Sigma_y$

[0135] A timestamp is tied to the aperture center in Equation 35. Other timestamp conventions must be converted explicitly. Sensor drift, saturation, cross-sensitivity, transport delay and missing samples belong to H_k and its calibration record. A raw imported value with a source label alone is not a calibrated physical observation.

Equation 36: an identifiable two-species optical embodiment

$$A_{10} = \log_{10}(I_0/I) = \ell\, c_T[\varepsilon_R(1-x) + \varepsilon_P x] + b + \varepsilon_{obs}$$

$$\hat{x} = (A_{10} - b - \ell\, c_T\, \varepsilon_R) / [\ell\, c_T(\varepsilon_P - \varepsilon_R)]$$

$\varepsilon_P \neq \varepsilon_R$; ℓ [m]; $\varepsilon_R, \varepsilon_P$ [$m^2\ mol^{-1}$] for this convention

[0136] This Beer-Lambert observation is an explicit calibrated embodiment for a sufficiently uniform optical sample over its verified linear range. Actual spectral coefficients, path length, blank, stray-light effects and other absorbing species must be measured. A multiwavelength observation $y = Hc + b + \varepsilon$ requires rank and conditioning adequate to distinguish the retained species.

Equation 37: exact aperture effect for the ideal trajectory

$$x_{avg}(t) = x_{eq} + (x_0 - x_{eq})e^{(-\lambda t)} \sinh(a)/a, \quad a = \lambda\Delta/2$$

$$\tau_c(x_{avg}) - t = -\ln[\sinh(a)/a]/\lambda \quad \text{when } \lambda_{ref} = \lambda$$

$$\text{small-aperture bias} \approx -\lambda\Delta^2/24; \quad \lim_{a\to 0} \sinh(a)/a = 1$$

[0137] Equation 37 assumes a centered aperture wholly within the same fixed-parameter segment, no latency, and a linear composition sensor. Its bias is deterministic measurement-model error if instantaneous inversion is used. Recording more timestamp digits does not remove this averaging effect.

21. Thermodynamic and kinetic closure

21.1 Nonideal activity and compatible reaction rates

Equation 38: species potentials, affinity and equilibrium

$$\mu_i = \mu_i^\circ(T,p) + RT \ln a_i; \quad a_i = \gamma_i\, c_i/c_i^\circ$$

$$A_r = -\Sigma_i\, \nu_{ir}\, \mu_i; \quad K_r = \exp[-\Delta_r G^\circ/(RT)]$$

equilibrium: $\prod_i a_i^{(\nu_{ir})} = K_r$

[0138] Activities are dimensionless and the concentration reference c_i° is retained. An ideal mole-fraction convention instead uses $a_i = \gamma_i x_i$. Activity coefficients must be compatible derivatives of a selected thermodynamic free-energy model; arbitrary independent γ_i values need not obey mixture consistency. The standard chemical potentials have a common reference convention. Only balanced differences enter reaction driving.

Equation 39: locally detailed-balanced kinetic embodiment

$$j_r = j_r^+ - j_r^-; \quad j_r^+/j_r^- = \exp[A_r/(RT)]$$
$$j_r = j_r^- \operatorname{expm1}[A_r/(RT)]$$
$j_r^- > 0$ is a separately identified kinetic scale [mol m^{-3} s^{-1}]

[0139] Equation 39 applies to a compatible elementary thermal reaction pair and its stated reservoirs. A catalyst, light-driven process, electrochemical drive or coarse-grained multistep mechanism requires the corresponding states and work terms. Local detailed balance relates rates and exchanged thermodynamic quantities under its assumptions; it is not an unrestricted identity for every measured net reaction rate [T15, T17]. A mechanism with several reactions carries a separate stoichiometric vector and affinity per elementary pair.

Equation 40: declared temperature dependence

$$k(T) = k(T_0) \exp[-E_a/R\ (1/T - 1/T_0)]$$
for the 1:1 retained convention: $k_f(T)/k_r(T)=K(T)$
$d \ln K/dT = \Delta_r H^\circ/(RT^2)$ for the retained standard-state model

[0140] An Arrhenius fit is a constitutive approximation over a measured temperature range. Independently fitting two rates without enforcing compatible equilibrium can violate the chosen thermodynamic model. For other stoichiometries, activity-normalized coefficients or the required reference-concentration factors enforce the equilibrium relation; raw dimensional rate constants of different orders cannot be equated to a dimensionless K. Pressure, ionic strength, catalyst state and stress may also affect kinetics; their dependencies are identified when material to the requested accuracy.

21.2 Generalized chemical time

Equation 41: a state-to-time map and variable-rate segment

$$\tau_c = H(z;\theta); \quad d\tau_c/dt = \nabla_z H \cdot F(z,u,t;\theta)$$
if H or θ changes explicitly: add $\partial H/\partial t + (\partial H/\partial \theta)\ \theta$
constant x_{eq}: $x - x_{eq} = (x_0 - x_{eq}) \exp[-\text{integral}_0^t \lambda(s)\, ds]$
fixed inverse gives $\tau_c = [\text{integral}_0^t \lambda(s)\, ds]/\lambda_{ref}$

[0141] A state-only inverse H exists only where the retained observation distinguishes progression along the intended trajectory. If temperature or forcing changes the trajectory, composition alone may not identify elapsed time. A multidimensional state observer or retained input history is then required. Cycles need phase and cycle count.

Retuning H after observing the validation residual cannot serve as independent evidence that the original clock was accurate.

[0142] Equation 17A remains the fixed ideal elementary-reaction specialization. For a nonideal reaction, the physical potentials and rates in Equation 39 are updated together and the inverse H must be recalibrated or derived for that model. Substituting a new μ into the old ideal inverse without updating its kinetics is inconsistent.

22. Species transport, flow and changing volumes

22.1 Local and moving-control-volume conservation

Equation 42: Eulerian and moving-domain species balances

$$\partial c_i/\partial t + \nabla\cdot N_i = \Sigma_r \nu_{ir} j_r + s_i; \quad N_i = c_i v + J_i$$
$$\frac{d}{dt}\int_{\Omega(t)} c_i \, dV = -\int_{\partial\Omega} (N_i - c_i w)\cdot n \, dA + \int_{\Omega} (\Sigma_r \nu_{ir} j_r + s_i) \, dV$$
$$M^{T}\nu = 0; \quad \text{elemental balance matrix B satisfies } B\nu = 0$$

[0143] Here v is mixture velocity and w is the control-boundary velocity. A registered moving specimen therefore cannot be treated as a fixed observation volume without accounting for relative transport. Species balances preserve elements and mass, not necessarily total mole number. Charge balance is also imposed for charged species.

Equation 43: well-mixed changing-volume compartment

$$\dot{n}_i = V \Sigma_r \nu_{ir} j_r + \dot{n}_{i,in} - \dot{n}_{i,out}; \quad c_i = n_i/V$$
$$\dot{c}_i = \dot{n}_i/V - c_i \dot{V}/V$$
$$x = n_P/n_T; \quad \dot{x} = (\dot{n}_P - x\,\dot{n}_T)/n_T$$

[0144] For an isolated one-to-one isomerization, n_T is constant. Dilution can cancel from the mole-fraction derivative even while concentrations, transport or non-first-order reaction rates change. The chosen physical variables determine which correction is required; a volume term is not added twice.

22.2 Constitutive transport and fluid closure

Equation 44: transport branches with declared validity

uniform T, neutral dilute branch: $J_i = -D_i c_i \nabla\mu_i/(RT)$

nonisothermal branch: $J = (J_e, J_1, \ldots) = L X$

$X = (\nabla(1/T), -\nabla(\mu_1/T), \ldots); \quad X^{T} L X \geq 0$

[0145] For the second branch, J_e is the nonconvective internal-energy flux including the energy carried by diffusing species; its entropy flux is $(J_e - \Sigma_i \mu_i J_i)/T$. The independent species-flux basis, reference frame and cross-coefficient units are retained. A positive-semidefinite symmetric dissipative part of L enforces nonnegative

transport entropy production. Neither arbitrary independent diffusivities nor a uniform-temperature law automatically close a nonisothermal multicomponent mixture.

Equation 45: optional incompressible Newtonian channel flow

$$\nabla\cdot v=0;\quad \rho(\partial v/\partial t+v\cdot\nabla v) = -\nabla p + \eta\nabla^2 v + \rho b$$

no-slip wall: $v=v_{wall}$; inlet/traction data and pressure gauge retained

[0146] Equation 45 is selected only for an incompressible Newtonian fluid with the stated density and viscosity assumptions. A porous homogenized model may replace it with a calibrated permeability law; pore-scale and homogenized laws are not applied simultaneously to the same unresolved volume. The equations require consistent inlet/outlet and wall conditions.

Equation 46: optional charged-species extension

$$\tilde{\mu}_i = \mu_i + z_i F_{const} \psi_e$$

$$-\nabla\cdot(\varepsilon_e \nabla\psi_e) = F_{const} \Sigma_i z_i c_i + \rho_{fixed}$$

use $\tilde{\mu}_i$ in ionic driving forces; retain electrical boundary work

[0147] Here F_{const} is the Faraday constant in C mol^{-1}, ψ_e is electric potential in V, ε_e is permittivity in F m^{-1}, and z_i is the signed charge number. This electrostatic branch requires permittivity, fixed charge, electrical boundary conditions and charge-conserving reaction stoichiometry. An electroneutral approximation is a separately justified scale limit. The neutral reference example does not require this branch.

23. Thermal, mechanical and structural realization

23.1 Temperature is a state or a measured maintained condition

Equation 47: closed-batch fixed-volume thermal closure

$$U=U(T,n,V);\quad C_V=(\partial U/\partial T)_{(n,V)}$$

$$C_V \dot{T} = \dot{Q}_{in} + \dot{W}_{nonflow} - \Sigma_i (\partial U/\partial n_i)_{(T,V)} \dot{n}_i$$

closed reaction: chemical term $= -V \Sigma_r \Delta_r U j_r$

lumped heat exchange example: $\dot{Q}_{in}=hA_s(T_{bath}-T)+P_{heat}$

[0148] The reaction internal-energy change in Equation 47 is derived from the same retained U and concentration convention. The closed-reaction specialization excludes feed/waste; an open compartment uses the stream enthalpy and non-flow work terms of Equation 23. Constant-pressure caloric closure uses enthalpy instead. Affinity is not inserted as an additional heat source on top of already counted chemical energy.

Equation 48: total control-volume energy when motion matters

$$E_{cv} = \int_\Omega \rho(e+|v|^2/2+\Phi_g)\, dV$$
$$\dot{E}_{cv} = Q_{in} + \dot{W}_{nonflow} + \Sigma_{in}\, \dot{m}(h+|v|^2/2+\Phi_g) - \Sigma_{out}\, \dot{m}(h+|v|^2/2+\Phi_g)$$

[0149] Non-flow work includes the selected mechanical/electrical work ports with a declared sign. Stream enthalpy already includes flow work. For a moving control surface the stream fluxes use relative normal velocity, and $\dot{W}_{nonflow}$ explicitly includes boundary pressure work $-\int_{\partial\Omega} p\, w\cdot n\, dA$ as well as the selected non-pressure work. Relative enthalpy flux alone is insufficient. Kinetic or gravitational terms can be dropped only after their contribution is bounded against the retained tolerance.

23.2 A concrete small-strain chemoelastic closure

Equation 49: one shared free energy in a reference volume

$$\psi_0 = \psi_{mix}(C,T)+\psi_{th}(T) + \tfrac{1}{2}(\varepsilon-\varepsilon^*) : C_{el} : (\varepsilon-\varepsilon^*)$$
$$\psi_{mix}=\Sigma_i\, C_i\{f_i^\circ(T)+RT[\ln(C_i/C_i^\circ)-1]\}+\psi_{ex}(C,T)$$
$$\varepsilon=\mathrm{sym}\nabla u; \quad \varepsilon^*=\alpha(T-T_0)I + \Sigma_i\, \beta_i(C_i-C_{i,0})$$
$$\sigma=\partial\psi_0/\partial\varepsilon; \quad \mu_i=\partial\psi_0/\partial C_i; \quad s_0=-\partial\psi_0/\partial T$$
$$e_0=\psi_0+Ts_0; \quad C_{V,0}=-T\,\partial^2\psi_0/\partial T^2 > 0 \text{ at fixed } C,\varepsilon$$

[0150] C_i is moles per reference volume; f_i° is a retained molar free-energy reference; ψ_0 is J m^{-3}; C_{el} is Pa; α has units K^{-1}; β_i is a strain-per-concentration tensor with units m^3 mol^{-1}. A composition-independent C_{el} gives the explicit chemical-mechanical term $-\sigma:\beta_i$ in μ_i. Composition-dependent elastic moduli add their derivative. Positive elastic stiffness, positive heat capacity, a permitted concentration domain and an appropriate mixture free energy are required. The matrix/solvent thermal term ψ_{th} supplies retained background caloric behavior, and U is the integral of e_0 with the chosen reference-volume convention. This is a constitutive model for its declared small-strain domain, not a measured law for every material.

Equation 50: force balance, boundaries and finite-strain alternative

$$\rho_0\, \ddot{u} = \nabla\cdot\sigma + \rho_0\, b; \quad u=u_D \text{ on } \Gamma_D; \quad \sigma n=t_N \text{ on } \Gamma_N$$
finite strain: $F_{def}=I+\mathrm{Grad}\, u$; $J_{det}=\det F_{def} > 0$; $c_i=C_i/J_{det}$
$$P=\partial\Psi_0/\partial F_{def}; \quad \sigma=J_{det}^{-1}\, P\, F_{def}^{T}; \quad \rho_0\, \ddot{u}=\mathrm{Div}\, P+\rho_0 b$$

[0151] The finite-strain branch requires an identified objective free-energy function Ψ_0 and the corresponding material flux and boundary formulation. Yield, creep, fracture, fatigue or viscoelastic response requires additional internal variables and evolution laws when those effects enter the intended operating range. Their material parameters cannot be inferred from lattice topology alone.

23.3 Actuator dynamics and geometry must share a mass balance

Equation 51: calibrated actuator and boundary motion

$$T_{act}\,\dot{a} + a = \mathrm{sat}[u_{cmd}(t-d_{act})]$$
$$q_{dep}(x,t) = \eta_{dep} \int K_{dep}(x,x')\, a_{dep}(x',t)\, dA'$$
$$v_n = v_s\cdot n + (q_{dep}-q_{rem})/\rho_s$$
$$\partial\varphi/\partial t + v_n|\nabla\varphi| = 0; \quad \dot{M}_s = \int_{\partial\Omega}(q_{dep}-q_{rem})dA$$

[0152] Equation 51 is a first-order actuator embodiment with calibrated delay d_{act}, response time T_{act}, deposition efficiency, normalized spatial kernel and saturation. q_{dep}, q_{rem} and the actuator mass-flux state a_{dep} are kg m^{-2} s^{-1}; ρ_s is kg m^{-3}. K_{dep} has units m^{-2} and integrates to one over the receiving surface x for each fixed source x′. The removal channel has its corresponding calibrated transfer law. Here v_s is the solid material velocity; a depositing boundary can move relative to that material. Define $M_s=\int_\Omega \rho_s\, dV$ and enforce interior continuity $\partial\rho_s/\partial t+\nabla\cdot(\rho_s v_s)=0$. The displayed mass-rate identity then follows with no other bulk source or mass port. The interface density is the locally retained solid density. Actual removal, deposition and reversible deformation select their applicable terms. The physical actuator transfer function is identified from command/response measurements; the current generated-register adapter does not identify it.

24. Initial/boundary conditions and deterministic solution

24.1 Data needed to solve the selected system

Subsystem	Required initial data	Required boundary/input data
Reference chemistry	Species amounts, temperature, volume, epoch and covariance.	Feed/waste, bath/heat input, permitted drive and timebase.
Spatial chemistry/flow	Concentrations, velocity or pressure-consistent initial state.	Impermeable walls or specified flux/chemical potential, inlet/outlet and pressure gauge.
Solid mechanics	Reference geometry, displacement, velocity and material/internal variables.	Displacement/traction partition, loads, contacts and constraints.
Thermal state	Temperature and material energy law.	Heat flux, bath or temperature data; electrical/mechanical power.
Actuation/observer	Actuator state, observer state, delays and controller state.	Commands, sampling schedule, saturation, calibration and noise model.

Subsystem	Required initial data	Required boundary/input data
LUT/geometry	Kernel, complete snapshot, schema, recipe and raw dependencies.	Canonical ordering, quantization, spatial frame, guards and execution environment.

24.2 A deterministic equation needs existence and uniqueness

Equation 52: sufficient finite-dimensional local condition

$$\dot{z}=F(z,t), \quad z(t_0)=z_0$$

$$||F(z_1,t)-F(z_2,t)|| \le L||z_1-z_2|| \text{ in the retained domain}$$

continuity in t and local Lipschitz continuity in z give local uniqueness

[0153] The ideal $R \rightleftharpoons P$ scalar ODE satisfies these conditions globally and preserves $0 \le x \le 1$ for positive rates. Global existence, positivity and constraint preservation must be checked for each extended system. A PDE requires appropriate regularity, ellipticity/dissipation, compatible boundaries and a solution concept. The architecture does not supply a universal uniqueness proof for arbitrary nonlinear multiphysics PDEs, contact, damage or topology change.

Equation 53: deterministic sensitivity bound

$$||\delta z(t)|| \le e^{(Lt)}||\delta z_0|| + \text{integral}_0^t e^{[L(t-s)]}\,||\delta F(s)||\,ds$$

[0154] Equation 53 is a conditional Gronwall bound for bounded perturbations under the declared Lipschitz condition. Determinism gives one trajectory for one exact model and initial state; it does not make that trajectory insensitive to uncertain inputs or omitted physics. Stiffness, instability and sensitive dependence can amplify small errors.

24.3 Hybrid events and reproducible digital execution

[0155] A physical controller combines continuous evolution and discrete decisions. Event guards, simultaneous-event priority, solver tolerances, saturation, command retries and minimum dwell time are part of the model. Positive dwell time or a proven event-separation condition prevents infinitely many controller transitions in a finite time interval. Observations and receipts remain recorded external inputs to replay.

Equation 54: a reproducibility contract

same retained bytes + K + recipe + numeric environment + event order
→ same reconstructed canonical table

source digest ≠ sensor authenticity ≠ physical correctness

[0156] The current Python serialization declares sorted ASCII keys, finite binary64 values and a fixed representation rule. Source files, dependency/library versions, numeric backend and platform constraints must be pinned if bitwise replay is required across machines. A digest detects changes relative to a trusted record; it neither authenticates the instrument nor proves that its reported physical value is true.

25. Stochastic chemistry and chemical-clock limits

25.1 A finite molecular population

Equation 55: reversible birth-death chemical master equation

$$a_+(m)=k_f(N-m);\quad a_-(m)=k_r\,m;\quad m=0,\ldots,N$$
$$P_m = k_f(N-m+1)P_{(m-1)} + k_r(m+1)P_{(m+1)} - [k_f(N-m)+k_r\,m]P_m$$
outside-state probabilities vanish; $\Sigma_m P_m=1$

[0157] This is the exact probability evolution of the stated well-mixed, independent-molecule continuous-time Markov model, with N conserved molecules and elementary first-order propensities. It is not an assertion that this Markov approximation resolves every microscopic physical effect. For linear propensities the mean fraction follows the same deterministic ODE, while individual count trajectories fluctuate [T14].

Equation 56: mean and variance for fixed initial count m_0

$$a=x_{eq}(1-e^{-\lambda t});\quad b=x_{eq}+(1-x_{eq})e^{-\lambda t}$$
$$E[m]=(N-m_0)a+m_0b$$
$$\mathrm{Var}(x)=[(N-m_0)a(1-a)+m_0b(1-b)]/N^2$$
equilibrium: $\mathrm{Var}(x)=x_{eq}(1-x_{eq})/N$

[0158] The variance in Equation 56 uses a fixed initial count m_0. A binomial $p(t)[1-p(t)]/N$ formula at all times would instead assume independently randomized initial states. A seeded simulated path is reproducible as a computation; the seed is not control over reaction events in an apparatus.

25.2 Stochastic inversion does not preserve an exact clock path

Equation 57: diffusion approximation and inverse-clock correction

$$dx=f(x)dt+b_N(x)dW;\quad b_N^2=[k_f(1-x)+k_r\,x]/N$$
$$dH=[H'f+\tfrac{1}{2}H''b_N^2]dt + H'b_N\,dW$$
$$H'=1/[\lambda_{ref}(x_{eq}-x)];\quad H''=1/[\lambda_{ref}(x_{eq}-x)^2]$$

[0159] Equation 57 is an Ito diffusion approximation for suitable large populations. It is stopped before leaving the retained inverse-clock branch, including approach to x_{eq}, as well as before molecular endpoints invalidate the approximation. It exhibits both an added drift and a stochastic term after nonlinear clock inversion. The exact molecular model has jumps, so an ordinary pathwise derivative $d\tau_c/dt$ is not available at each reaction event. Equation 17A describes the declared deterministic reaction trajectory; it cannot remove physical reaction fluctuations by renaming the variables.

25.3 Finite useful duration of a relaxation clock

Equation 58: composition-resolution bound and usable interval

u_x contribution: $u(\tau_c) \approx u_x/[\lambda_{ref}|x_{eq}-x|]$
ideal $\lambda_{ref}=\lambda$: $u(\tau_c)=u_x e^{(\lambda t)}/[\lambda|x_{eq}-x_0|]$
$t_{max} \le (1/\lambda) \ln[\lambda|x_{eq}-x_0|u_{max}/u_x]$

[0160] The final bound uses constant u_x, exact retained parameters and first-order propagation; all additional uncertainty reduces or changes the usable interval. If the logarithm argument is at most one, the required uncertainty is not met for any positive interval under those assumptions. At equilibrium the state-only relaxation clock loses time information. A measured renewal or a properly identified driven oscillator supplies subsequent timing segments.

26. Uncertainty, identifiability and independent validation

26.1 Full propagation through the chemical-time inverse

Equation 59: clock Jacobian on one fixed monotone segment

$H=-(1/\lambda) \ln[(x_{eq}-x)/(x_{eq}-x_0)]$
$H_x=1/[\lambda(x_{eq}-x)]$; $H_{x_0}=-1/[\lambda(x_{eq}-x_0)]$
$H_{x_{eq}}=\{1/(x_{eq}-x_0)-1/(x_{eq}-x)\}/\lambda$; $H_\lambda=-H/\lambda$
$u^2(H)=J_H \Sigma_{(x,x_0,x_{eq},\lambda)} J_H^T$

[0161] The covariance includes calibration/observation correlation. If k_f and k_r are the calibrated parameters, transform their covariance jointly into λ and x_{eq} or differentiate in the original parameterization. A first-order covariance approximation can fail near the inverse singularity or with appreciable probability outside the valid branch; retained-distribution propagation and explicit invalid-domain probability are then required [T11, T12]. Model discrepancy is evaluated separately or included through a justified discrepancy model.

Equation 60: residual dependence and admissibility

$r=y_{obs}-y_{pred}$; $\Sigma_r=J_r \Sigma_{inputs} J_r^T$
$q=r^T \Sigma_{r^+} r$ on an identified independent residual subspace
also require the residual component outside that subspace to vanish
report rank, assumptions, coverage and fixed acceptance thresholds

[0162] A chi-squared interpretation of q requires its distributional and parameter-estimation assumptions; it is not supplied by the quadratic form alone. Algebraically duplicated residuals are not counted as independent checks. A pseudoinverse must not silently discard a residual component that the model says has zero variance. Simple engineering bounds may be used instead, provided the tolerance and uncertainty interpretation are explicit.

26.2 Quantifying accuracy requires a validation target

Equation 61: a physical accuracy statement with a domain

for t and conditions in D: $\Pr(|\hat{t}_c - t| \le E_t) \ge p_{cov}$
for geometry in D: $\Pr(d_H(\partial\Omega_{built}, \partial\Omega_{target}) \le E_G) \ge p_{cov}$
state D, confidence/evidence, error budget and calibration version

[0163] E_t and E_G are dimensional error limits; p_{cov} is the declared coverage probability under its statistical model. A percentage such as physically exact to 99.9% has no determinate meaning without the measurand, scale, interval, operating domain and evidence. No calibrated real-apparatus error distribution is supplied with the current package, so a numerical physical accuracy rating cannot be inferred from its test count.

26.3 Calibration and validation must be separated

[0164] Fit the reaction, sensor and actuator parameters on a recorded calibration dataset; retain their covariance and residual structure. Freeze those parameters and policy thresholds before predicting a separate validation interval. Use independently acquired atomic timestamps and chemical measurements. Challenge the model at multiple temperatures, concentrations, delays and geometries inside the intended range. Report failed predictions as well as admitted runs.

[0165] An energetic measurement or independently characterized rate channel can add evidence for the affinity relation. Computing μ_R, μ_P, j_+ and j_- from the same x, k_f and k_r only checks an algebraic consistency relation. The source-dependence flags in the implementation are necessary, but actual independence comes from acquisition and calibration design.

27. SDF, numerical geometry and LUT approximation

27.1 Physical predicate uncertainty becomes boundary uncertainty

Equation 62: local boundary sensitivity and gate bounds

$g(z(x))=0; \quad |\delta n| \approx |\delta g|/||\nabla g||$ where $||\nabla g||>0$
$|\delta A| \le \min[1, |\delta\varphi_A|/(2\varepsilon_A)]$ for the clipped linear gate
$|\delta S| \le |\delta A|+|\delta B|+|\delta C|, \quad S=ABC$ with all gates in [0,1]

[0166] The local normal-displacement approximation requires a regular boundary and small perturbations. Registration, sensor calibration and field reconstruction contribute to $\delta\varphi$. Near a flat or critical predicate, a small physical-value error can produce a large or nonunique geometric change. Renaming a scalar physical field as a signed distance does not provide a metric error bound; distance reconstruction and its geometry must be retained.

Equation 63: robust threshold membership

certain selected: $\hat{S}-\varepsilon_S \geq \hat{\tau}_g+\varepsilon_\tau$
certain unselected: $\hat{S}+\varepsilon_S < \hat{\tau}_g-\varepsilon_\tau$
otherwise: ambiguous membership → hold, refine or remeasure

[0167] This is an interval-bound rule, not a confidence statement unless the intervals have that interpretation. Protected regions are expanded by their registration/geometry uncertainty before a destructive operation. A membership ambiguity is not resolved by choosing a convenient rounding direction.

27.2 Grid resolution and metric reconstruction

Equation 64: a cell certificate for a true 1-Lipschitz SDF

$r_{cell}=\frac{1}{2}\sqrt{h_x^2+h_y^2+h_z^2}$
$\varphi(\text{center})+r_{cell} \leq 0 \Rightarrow$ the whole cell is inside
$\varphi(\text{center})-r_{cell} > 0 \Rightarrow$ the whole cell is outside
other cells intersect the possible boundary band

[0168] A cell-center sign rule can miss thin components, narrow passages or protected features. A blanket half-diagonal Hausdorff error claim would be invalid without resolution and regularity conditions. A nonmetric implicit field uses its actual Lipschitz bound, not the SDF bound by assertion. Mesh convergence, minimum-feature checks, topology and process-resolution limits accompany the chosen reconstruction.

27.3 LUT quantization and interpolation error

Equation 65: deterministic lookup approximation bounds

$\|\pi(z)-\pi(Qz)\| \leq L_\pi \|z-Qz\|$ for a Lipschitz policy
1D linear interpolation: $|f-f_{LUT}| \leq (\Delta z^2/8) \max|f''|$
hold-time state change $\leq \sup\|F\| \Delta t$ within the retained domain

[0169] Equation 65 requires the stated regularity. A stepwise threshold policy is discontinuous at its boundaries and does not possess a finite global L_π there. Its uncertainty is managed with explicit interval decisions, hysteresis and bounded transitions. Exact replay of a quantized table is compatible with nonzero approximation error relative to a continuous physical controller.

28. Control stability and physical regeneration

28.1 A physical controller needs a dynamics and delay argument

Equation 66: a local sampled feedback test

$$\delta z_{n+1}=A_d\,\delta z_n+B_d\,\delta u_n;\quad \delta u_n=-K\,\delta z_n$$

local linear stability requires spectral radius$(A_d-B_d K)<1$

include actuator/observer/delay states in A_d and B_d

[0170] The condition is local to the identified operating point and linearization. Saturation, discrete topology changes, irreversible removal and large parameter changes require separate bounds or reachability checks. The current five-level threshold schedule is an implemented design policy; the package contains no identified physical plant matrices proving its closed-loop stability.

Equation 67: hysteresis, dwell and physical admissibility

switch up only beyond q_{high}; switch down only below q_{low}

$q_{low} < q_{high};\quad \Delta t_{dwell} > 0$

irreversible removal: $\Omega_{next} \subseteq \Omega_{current}$ unless deposition is executed

admit action only if resource, load, geometry and uncertainty bounds hold

[0171] Threshold switching must account for the actual direction of material change. A later software state cannot restore removed matter. A failed post-observation retains the changed plant state and its evidence even when it leaves the active LUT unchanged. That rule is already present in the generated-register demonstration and becomes a physical recovery requirement for an apparatus.

28.2 Resource closure of regeneration

Equation 68: renewal inventory and continued operation

$$\dot{R}_j = \text{supply}_j - \text{consumption}_j - \text{waste}_j;\quad R_j \geq 0$$

renewal complete $\Leftrightarrow$ measured restored state $\in$ admitted restart set

$$\tau_{global} = \Sigma \text{ completed chemical segments} + \Sigma \text{ declared bridge durations} + \text{active chemical segment}$$

[0172] Resource entries have physical units: mol or kg for matter, J for energy, and identified spare components where needed. Renewal includes pumps, heaters, separation, sensing, actuation and computation within the declared system boundary. An external power or feed supply is an explicit port. Software regeneration is demonstrated; physical replacement/recharge and continued autonomous operation require observed completed cycles.

[0173] The retained interpreter, sensors, actuators, power supply and stored raw evidence are dependencies of the implemented regeneration. The term autopoietic is used operationally for renewal of the declared organization and, in a physical

embodiment, the specified components. It does not establish that all hardware can manufacture itself or that resources are unbounded.

29. Quantitative exactness audit

29.1 What the numerical experiments measure

[0174] The separate audit module applies specified perturbations to the same reference reaction and to the sampled SDF lattice. Its outputs are synthetic diagnostics and analytic calculations with retained assumptions. They measure sensitivity, conditioning and implementation consequences; they are not observations from an experimental apparatus. The original reference run remains preserved separately.

Audit item	Recorded outcome
Reference implementation verification	24 tests passed; six LUT generations; five commits.
New quantitative invariants	11 tests passed, zero failures or errors.
Repeated audit execution	11 report/data files reproduced byte for byte in the recorded runtime.
Independent physical observations	0 supplied in these execution records.
Clock uncertainty cutoff	$u_x=10^{-5}$ alone reaches $u(\tau_c)=0.10$ s at 139.017041 s.
Dependent residuals	The inspected first-order residual covariance has rank 1; duplicate derived channels are not independent evidence.

29.2 Clock and thermodynamic sensitivity

Declared diagnostic	Calculated result	Meaning
$u_x=10^{-5}$; t=60 s	$u(\tau_c)=0.00423968$ s	First-order composition-readout contribution only.
$u_x=10^{-5}$; t=120 s	$u(\tau_c)=0.04673478$ s	The same readout precision loses time resolution near equilibrium.
Common rate error ±1%; t=60 s	Clock bias ±0.600000 s	Equilibrium unchanged; nominal inverse uses the wrong kinetic scale.
Centered 0.1 s sensor aperture	Bias -16.666664 μs	Applying an instantaneous inverse to an averaged composition.
Fixed $N=10^6$ molecules; t=60 s	Clock SD ≈ 0.19556184 s	Intrinsic count noise; linearized inverse under the specified Markov model.
Assigned $E_a=50$ kJ/mol; $\Delta T=+0.01$ K	Bias at 60 s: +0.04060222 s	Illustrative equal-barrier Arrhenius scenario, not a measured activation energy.
Assigned $\gamma_P/\gamma_R=1.2$; x=0.73	Affinity 257.786031 → -194.181274 J/mol	The compatible net reaction flux changes sign; potentials and kinetics must change together.

[0175] A shared 1% scale error in both reaction rates preserves the equilibrium fraction but changes the inferred elapsed-time scale by 1% when the inverse keeps its nominal rate. A finite integration aperture gives a different error mechanism even with exact nominal rates. Nonideal activity corrections modify affinity, and compatible kinetics must then change with the same thermodynamic model. These examples identify physical quantities that a successful internal replay cannot estimate by itself.

29.3 Spatial uncertainty and admitted evidence

Grid spacing (mm)	Removed volume (mm^3)	Difference from finest tested grid (mm^3)
1	316.00000000	143.21484375
0.5	228.50000000	55.71484375
0.25	170.12500000	-2.66015625
0.125	166.64843750	-6.13671875
0.0625	172.78515625	0.00000000

[0176] The 0.5 mm grid differs from the 0.0625 mm comparison by 55.71484375 mm^3 (32.245156% of the finer-grid value). Refinement is nonmonotone, so the finest tested result is not a certified continuum volume. At the original grid, prescribed threshold

±0.01 and window-registration shifts ±0.1 mm produce 226.0-252.5 mm^3. Every changed cell lies in the derived 26.5 mm^3 uncertainty band (212 cells).

[0177] A simple cylinder cross-section explains part of the resolution effect: radius 0.8 mm has analytic area 2.01061930 mm^2. At 0.5 mm spacing, its 12 selected midpoint cells represent 3.00000000 mm^2, a 49.207759% excess for that primitive. The selected full lattice has additional intersections and window cuts; the primitive comparison is not a correction factor for its volume.

[0178] A change in selected voxels is a sensitivity result for the retained geometry and perturbation, not a fabrication tolerance. The audit also records the mathematical dependence of the ideal affinity residual on the same composition/model inputs. The appropriate validation result is an independently measured error distribution and usable operating domain, not the number of passed software tests.

30. Completion register and experimental decision criteria

30.1 Mathematical closure added by this edition

[0179] The added equations close the architectural categories of timebase, observation, thermodynamics, kinetics, transport, thermal state, mechanics, actuation, initial/boundary conditions, stochastic behavior, uncertainty, numerical geometry and control. They form selectable consistent embodiments. A physical instance selects one compatible branch per process and supplies its measured coefficients; mutually exclusive branches are not stacked indiscriminately.

ID	Remaining empirical item	Current state	Required closure evidence
G01	Reaction identity and mechanism	R and P are abstract species; numerical rates are stipulated.	Select actual species, solvent/phase and mechanism; independently estimate rates, equilibrium and their covariance.
G02	Atomic realization	Generated Cs-equivalent ticks represent an ideal reference.	Supply clock type, actual output ratio, frequency correction, lock history, delay and uncertainty.
G03	Chemical sensing	Composition is analytic-generated in the delivered example.	Provide sensor response, raw measurements, cross-sensitivity, aperture, calibration and drift records.
G04	Thermodynamic kinetics	Ideal activities and compatible standard potentials close the chosen model.	Identify activity coefficients and kinetic prefactors; check detailed-balance assumptions for every reaction step.
G05	Spatial and thermal closure	The architecture names transport and energy ports; real coefficients are absent.	Supply diffusion/mobility, flow, thermal properties, geometry, boundary conditions and initial state.

ID	Remaining empirical item	Current state	Required closure evidence
G06	Mechanical and actuator closure	Thresholds select a designed lattice; the actuator is a generated register.	Identify material law, loads, fabrication transfer function, mass flow, delay, rate limits and observed geometry.
G07	Independent evidence	Several residuals reuse the same composition and calibrated model.	Retain dependence graphs, full covariance and separate calibration and validation intervals.
G08	Clock conditioning	The relaxation inverse becomes singular at equilibrium.	Use an uncertainty-based usable interval and measured recharge or an identified oscillator phase observer.
G09	Closed-loop robustness	The five threshold levels are an engineering policy.	Supply stability or reachability analysis, hysteresis, delays, protected-set margins and irreversible-action rules.
G10	Stochastic and numerical error	Bitwise records and mesh closure do not bound physical error.	Characterize finite-population/sensor noise, model discrepancy, spatial discretization and process tolerance.
G11	Physical regeneration	The table can rebuild itself from retained dependencies.	Demonstrate replenishment/repair with measured matter, energy, completion and continued operation.

30.2 A concrete validation sequence

Stage	Experiment or check	Decision record
1. Identify	Choose actual R/P chemistry, phase, material, sensor, clock and actuator; define operating ranges.	Completed physical-instance manifest with units and source identities.
2. Calibrate	Measure sensor response, kinetics/equilibrium, clock transfer and actuator dynamics.	Parameter estimates, covariance, residuals and validity ranges.
3. Freeze	Fix model, controller, tolerances, initial-state protocol and calibration version.	Immutable validation configuration.
4. Predict	Predict separately acquired composition, time, energy and structural results.	Predictions retained before the corresponding measurements.
5. Challenge	Vary declared temperatures, load, delay, composition and geometry; include fault cases.	Coverage, bias, residual structure, failures and domain limits.
6. Renew	Run measured reset/repair cycles with matter/energy accounting and post-action sensing.	Physical restoration and continued operation evidence.

30.3 Present verdict

[0180] The architecture is mathematically coherent as a typed geometric/controller construction and as the specified ideal reversible-reaction example. Its scalar ideal dynamics and retained LUT reconstruction are deterministic under their stated inputs and environment. Equation 17A is conditionally exact mathematics. The supplied evidence does not establish a physically exact apparatus, a numerical real-world accuracy bound, general closed-loop stability, material performance or physical self-production.

[0181] The principal flaws arise if a conditional identity is treated as experimental confirmation, dependent calculations as independent measurements, ideal kinetic parameters as universal material constants, a mesh as a fabricated validated structure, or a digest/replay result as evidence of physical truth. Version 3 makes those distinctions explicit and supplies the equations and evidence contracts required to test a concrete physical realization.

Claims

What is claimed is:

1. A computer-implemented method for generating a lattice-bearing structural geometry, comprising: obtaining, by one or more processors, a first signed distance field representing a biochemical operating window and a second signed distance field representing a biomechanical operating window in a common physical coordinate frame, each signed distance field using a negative-interior convention; mapping the first and second signed distance fields to respective first and second dimensionless gate fields; obtaining a structural modulation field defined in the common physical coordinate frame; evaluating a first Hadamard product of the first and second gate fields and a second Hadamard product of the first product and the structural modulation field to obtain a coupled scalar field; evaluating a lattice-generating signed field; constructing a negative solid from a spatial intersection of a region of the lattice-generating signed field and a region selected according to the coupled scalar field; and evaluating a constructive solid geometry Boolean difference that subtracts the negative solid from a base solid to generate the structural geometry.

2. The method of claim 1, wherein at least one operating window is constructed by applying a specified physical-domain predicate to a registered biochemical or biomechanical quantity and constructing a signed distance field of the resulting operating set.

3. The method of claim 1, wherein mapping a signed distance field to its gate field comprises applying a hard membership map or a bounded transition map having a transition width expressed in physical length units, and wherein each gate field has values in the interval from zero to one.

4. The method of claim 2, wherein the biomechanical operating window is constructed from a biomechanical quantity obtained by scalarizing a mechanical tensor or combining a plurality of load cases, and wherein that operating window identifies a region satisfying a specified mechanical removal-permission condition.

5. The method of claim 1, wherein spatial registration verifies an origin, a sample convention, a grid spacing, a direction matrix, and a length unit, and explicitly resamples an input field when its physical sample positions differ from those of the common frame.

6. The method of claim 1, wherein the structural modulation field is a binary eligibility field or a bounded continuous field and remains active in an exterior collar used to extend a channel cutter through a boundary of the base solid.

7. The method of claim 1, wherein the selected region comprises positions at which the coupled scalar field is greater than or equal to a positive threshold no larger than one, and wherein a region equal to the threshold is resolved by constructing selected-region membership before reconstructing its boundary.

8. The method of claim 1, wherein the lattice-generating signed field is formed by a pointwise minimum of signed primitive fields placed according to a spatial repetition rule or a centerline network, the primitive fields representing cylindrical channels, capsule channels, struts, or sheets.

9. The method of claim 1, further comprising using the coupled scalar field to set a spatially varying radius or thickness of the lattice between specified bounds before constructing the negative solid.

10. The method of claim 1, wherein signed-field intersection uses a pointwise maximum, signed-field union uses a pointwise minimum, and signed-field difference uses a pointwise maximum of a first field and the negative of a second field, with a distance-reconstruction operation applied when a subsequent operation requires Euclidean-distance values.

11. The method of claim 1, wherein constructing the negative solid excludes positions designated as protected by a protected-region field, and wherein the difference and any subsequent remeshing preserve the designated protected region within the base solid.

12. The method of claim 1, wherein the structural geometry is generated as a union of retained physical grid cells and its boundary is formed by triangulating exposed faces while omitting shared interior faces, or is generated by extracting a boundary from an interpolated signed field with a recorded sample and isovalue convention.

13. The method of claim 1, further comprising generating a geometry record that includes coordinate units, lattice parameters, gate maps, coupling parameters, ordered Boolean operands, representation type, geometric volume, and boundary-topology information.

14. The method of claim 1, further comprising using the generated structural geometry to update at least one domain quantity, reconstructing an operating-window field from the updated quantity, and repeating the coupling and geometry generation according to a recorded termination condition.

15. The method of claim 1, further comprising deriving build instructions from the generated structural geometry and controlling an additive manufacturing apparatus according to the build instructions to fabricate a physical structural component.

16. A computing system for generating a lattice-bearing structural geometry, comprising one or more processors and memory storing instructions that configure the system to: maintain spatially registered biochemical and biomechanical signed-distance window records; map the window records to first and second bounded gate fields; evaluate a first elementwise product of the gate fields and a second elementwise product of the first product and a structural modulation field; generate a lattice signed field; construct a negative solid selected jointly by the lattice signed field and the resulting coupled scalar field; and evaluate an ordered Boolean difference that subtracts the negative solid from a base solid to output the structural geometry.

17. The computing system of claim 16, wherein the memory stores typed field records that distinguish signed distance, signed implicit geometry, dimensionless gate, coupled score, and membership mask, and wherein an operation requiring metric distances obtains a distance-calibrated field from the represented boundary.

18. A non-transitory computer-readable storage medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform the method of claim 1.

19. A computer-implemented method for generating a surface-lattice structural geometry, comprising: obtaining three spatially registered signed geometry fields representing biochemical, biomechanical, and structural boundary families; normalizing the fields by respective positive reference lengths; evaluating a first Hadamard product of two normalized fields and a second Hadamard product of the first product and the third normalized field; constructing a volumetric band selected by the magnitude of the resulting signed product; and evaluating a regularized Boolean difference that subtracts a negative solid comprising the volumetric band from a base solid to generate the structural geometry.

20. The method of claim 19, wherein the constituent zero surfaces are used to construct a distance-to-surface-union field, and wherein the volumetric band is further constrained by a prescribed physical-distance band selected from the distance-to-surface-union field.

21. An apparatus for regenerative control of a physical lattice structure, comprising: an atomic time-reference assembly; a chemical reaction compartment; a sensor configured to measure a chemical state of the compartment; a structural actuator; and one or more processors with memory storing a self-describing lookup table, wherein the processors are configured to obtain atomically referenced elapsed time, infer chemical elapsed time from the measured chemical state using a reaction calibration, determine reactant and product chemical-potential sums for the reaction, evaluate clock, kinetic and thermodynamic consistency conditions for the two elapsed times and two potential sums, regenerate entries of the lookup table from an admitted physical observation and a regeneration recipe identified by the table, generate a structural target using spatially registered signed-distance fields and two successive Hadamard products of bounded field operands, issue a command to the structural actuator according to the target, and commit a successor table generation after obtaining a receipt and a required post-action physical observation.

22. The apparatus of claim 21, wherein a physical binding comprises an ordered tuple of the atomically referenced elapsed time in seconds, the chemically inferred elapsed time in seconds, the stoichiometric reactant chemical-potential sum in joules per mole of reaction, and the stoichiometric product chemical-potential sum in joules per mole of reaction, with the reaction stoichiometry, standard states, calibration and observation epoch retained in a binding record.

23. The apparatus of claim 22, wherein the chemical reaction compartment implements a calibrated reversible first-order reaction, chemical elapsed time is obtained by

logarithmically inverting a measured displacement from equilibrium, and the difference between the reactant and product chemical-potential sums equals RT times the logarithm of a ratio j_{+}/j_{-} of forward and reverse molar reaction fluxes for the elementary reaction under local detailed balance, where R is the molar gas constant and T is thermodynamic temperature.

24. The apparatus of claim 23, wherein the chemical reaction compartment serves as a reference compartment that maintains the boundary conditions of the retained clock calibration independently of a changing volume or flow condition of the physical lattice structure, and the table retains a declared sampling, transport or control relationship between the reference compartment and spatial observations of the structure.

25. The apparatus of claim 21, wherein the lookup table includes meta-rows describing a channel registry, physical units, a schema, a regeneration recipe and complete reconstruction dependencies, and wherein regeneration reconstructs working rows and predictive rows from the meta-rows and retained physical observations.

26. The apparatus of claim 25, wherein a table generation is represented by a canonical body and a verified envelope, the body contains a symbolic current-body reference and a predecessor digest, a current-body digest is computed over the canonical body and stored in the envelope outside that body, and the symbolic current-body reference resolves through the verified envelope.

27. The apparatus of claim 21, further comprising material or energy interfaces configured to renew a chemical-clock segment or a physical component, wherein the successor table generation records supplied and removed material, a restored measured condition, an observation epoch and an identity of the renewed component.

28. The apparatus of claim 21, wherein the required post-action physical observation comprises a registered measurement of the structure and the processors evaluate a geometric discrepancy, a protected-region condition and a material-balance residual before admitting the realized structural target.

29. A method of operating a regenerative physical lattice apparatus, comprising: acquiring an atomic-reference observation and a chemical-state observation; calculating a first elapsed time from the atomic-reference observation and a second elapsed time from the chemical-state observation under a retained calibration; calculating reactant and product chemical-potential sums for a declared reaction; evaluating clock, kinetic and thermodynamic consistency conditions; using an admitted physical state to regenerate a self-describing lookup table and select a bounded lattice-construction or actuation parameter; generating a negative solid through field coupling and lattice selection; subtracting the negative solid from a base solid to define an actuator target; issuing an identified physical command to an actuator according to the target; obtaining a receipt for that command and a subsequent physical observation; and activating a successor table body linked to the preceding body when the subsequent observation meets a retained physical closure criterion.

30. The method of claim 29, wherein the bounded parameter is a dimensionless coupled-field threshold, a channel radius, a sheet thickness or a material actuation dose selected by explicit intervals of measured chemical progress, and wherein the parameter mapping is stored separately from the reaction constitutive law.

31. The method of claim 29, wherein a chemical-time uncertainty is propagated from composition and calibration covariance, and wherein a measured state near equilibrium, outside a monotone clock segment, with an unresolved calibration reference, or beyond a retained residual budget prevents activation of the successor table.

32. The method of claim 29, further comprising physically restoring a measured initial chemical condition and establishing a new chemical-clock segment, with outgoing and incoming observations, renewal resource flows and any atomic bridging interval retained in an epoch record.

33. The method of claim 29, wherein the second elapsed time is obtained from an unwrapped measured phase of a chemical oscillator using a retained reference period, cycle number, crossing rule and phase calibration, and wherein missing or ambiguous cycles are identified in the observation record and prevent activation based on the affected interval.

34. The method of claim 29, wherein the physical-state update includes mass-balanced reaction and transport equations and a material or energy balance, and wherein a biochemical signed-distance operating window is reconstructed from an activity, reaction-affinity or timing-consistency predicate on the updated physical state.

35. The method of claim 29, further comprising restoring a table generation by verifying a retained complete snapshot, resolving its reconstruction dependencies, replaying its regeneration recipe with retained observations and receipts, and comparing the resulting canonical body and geometry identity with the saved generation.

36. A non-transitory computer-readable storage medium storing instructions that, when executed by processors connected to atomic-reference, chemical-sensing and actuation interfaces of a physical apparatus, cause the apparatus to perform the method of claim 29.

Abstract

TOWNSENDCAPSTONECORNER

A regenerative physical architecture couples atomic-reference timing, a chemical clock, thermodynamic state and signed-distance-field lattice synthesis. An ordered binding comprises atomically referenced elapsed time, chemically inferred elapsed time, and reactant and product chemical-potential sums. A reaction model relates these quantities through kinetics, affinity and physical balances. Admitted observations regenerate a self-describing lookup table containing physical entries, predictive entries, control entries, schema information and a regeneration recipe. Each generation retains complete reconstruction dependencies and a finite self-reference linked to its predecessor. Bounded table outputs control spatially registered biochemical and biomechanical windows or lattice parameters. Two successive Hadamard products couple bounded fields to select a lattice negative solid, which is subtracted from a base solid. An apparatus records an actuator command, receipt and subsequent physical observation before activating a successor generation. Embodiments include chemical-clock renewal, structural fabrication, spatial transport coupling and measured geometric feedback.

Representative drawing: FIG. 7.

FIG. 1. SDF system architecture

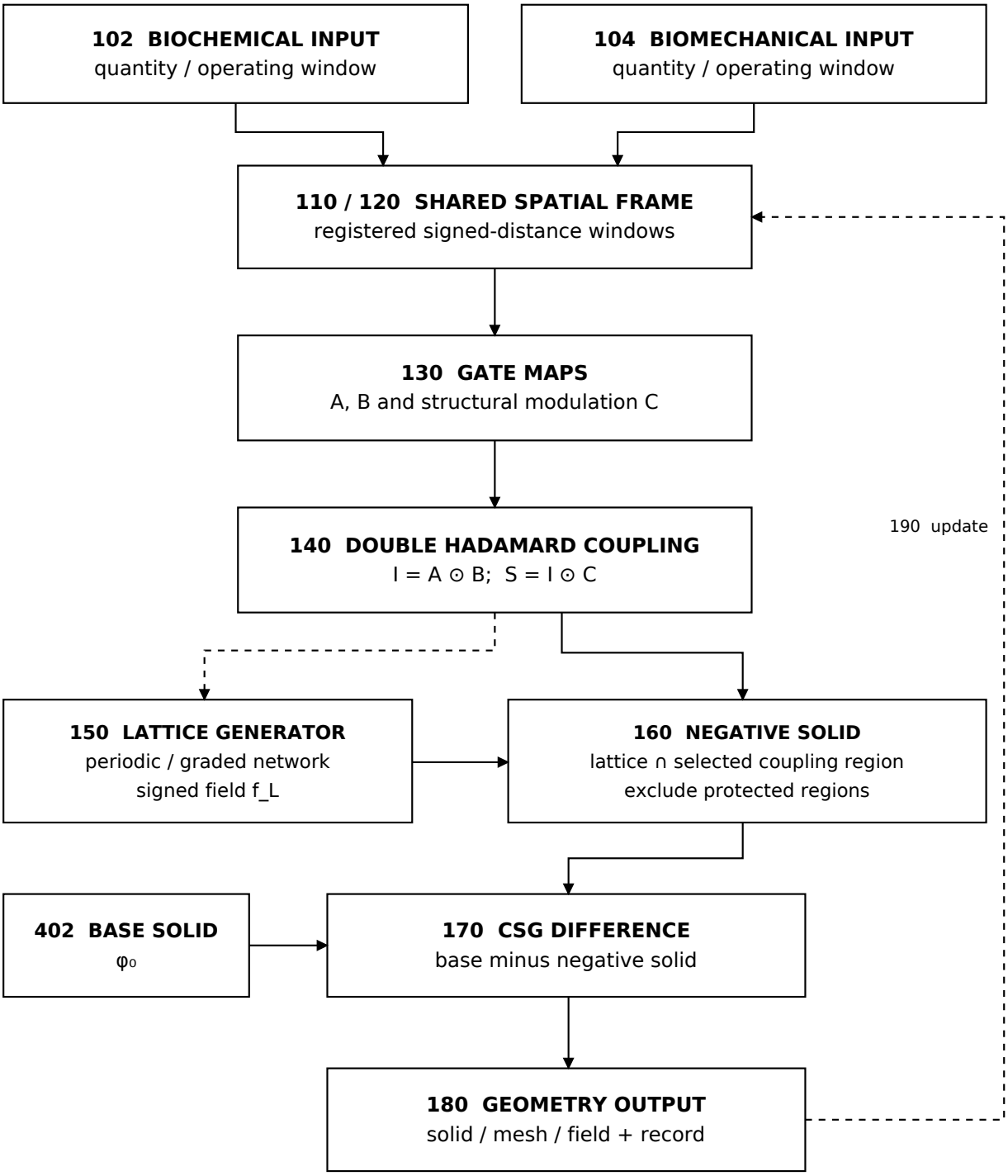


FIG. 1. Domain windows, SDFs, gate maps, double Hadamard coupling, lattice generation, and ordered solid subtraction. Dashed connections show optional parameter modulation and geometry-driven updates.

FIG. 2. Typed field algebra

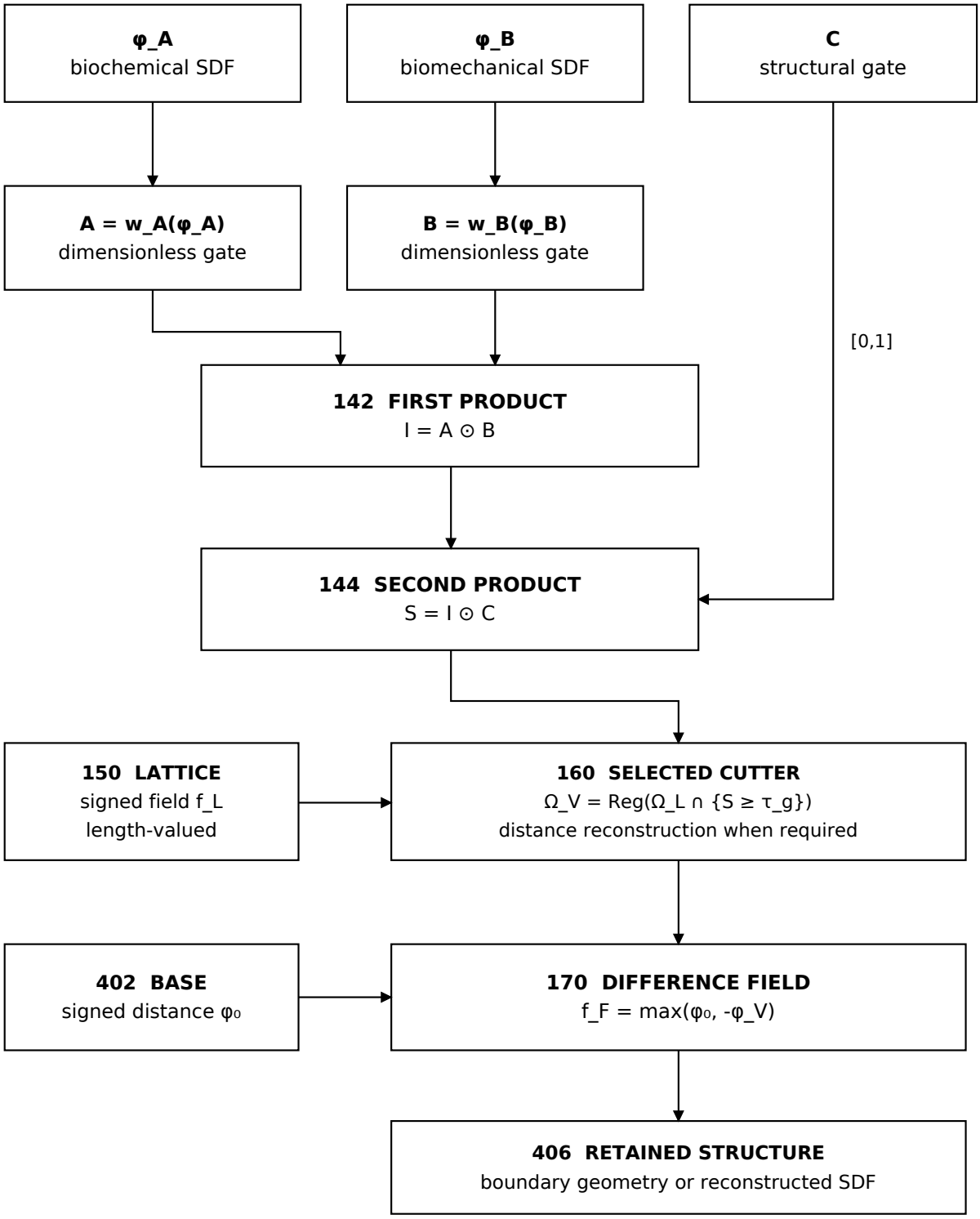
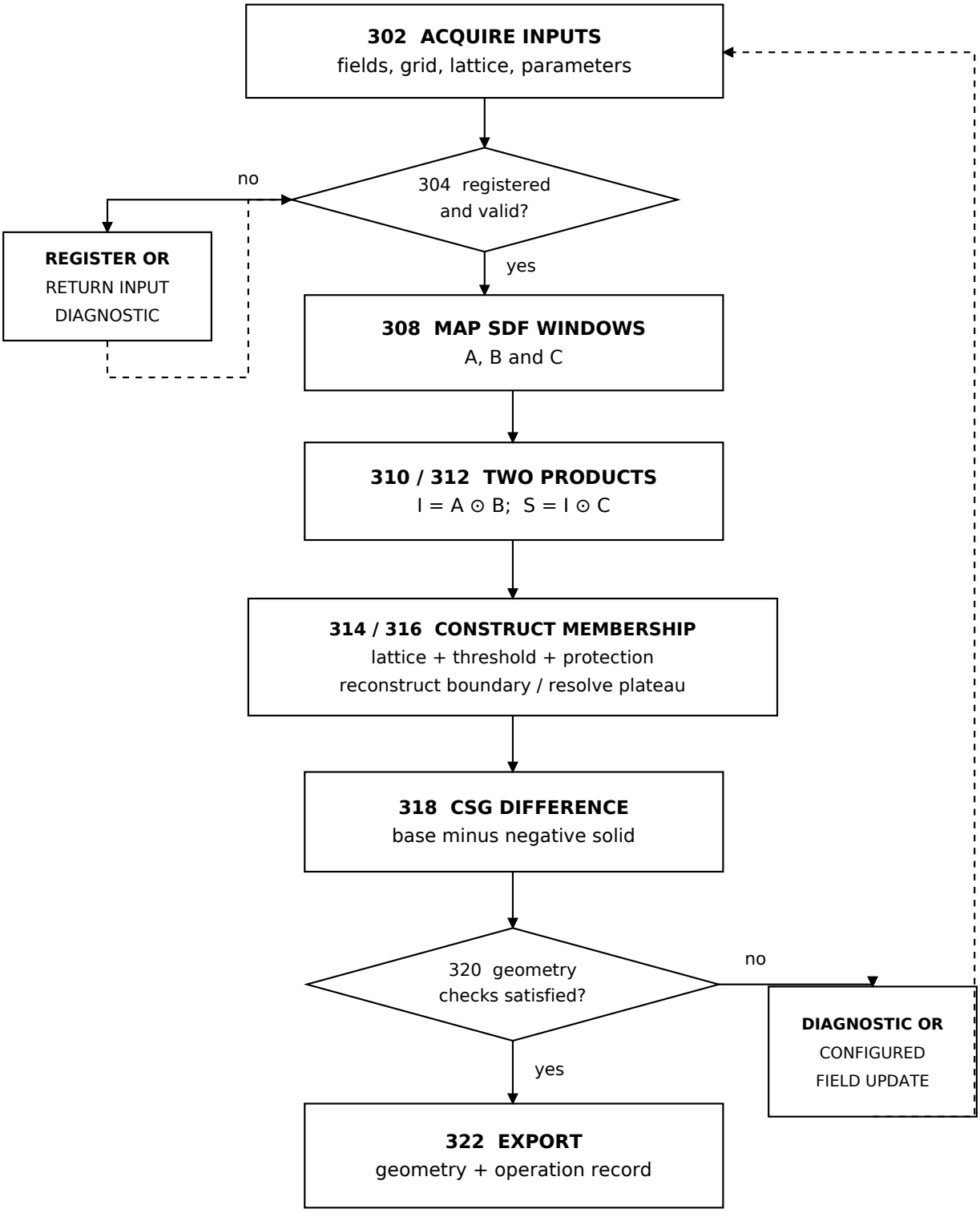


FIG. 2. Signed-distance fields retain geometric meaning. Dimensionless gates are multiplied in two stages. The selected coupled region and lattice define the negative solid used in the difference.

FIG. 3. SDF execution flow



Dashed paths: explicit correction or configured iteration.

FIG. 3. Defined input, membership, boundary, and output stages. Registration correction and geometry-driven updates follow the explicit execution configuration.

FIG. 4. CSG lattice construction

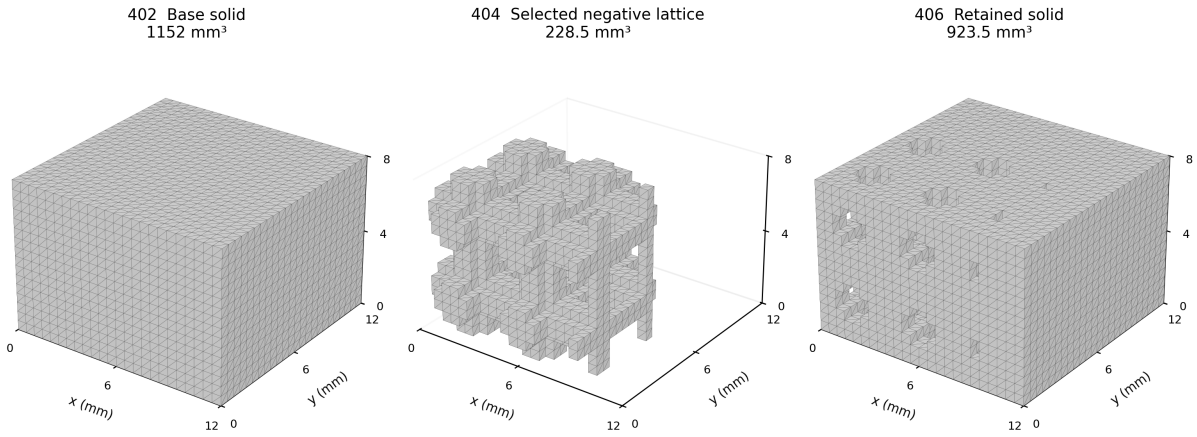


FIG. 4. Cell-union surfaces from the SDF lattice example. Base volume: 1,152 mm³. Selected negative volume: 228.5 mm³. Retained volume: 923.5 mm³. All panels use the same coordinates and orientation.

VOLUME PARTITION

1,152 mm³ = 228.5 mm³ + 923.5 mm³

9,216 cells = 1,828 removed cells + 7,388 retained cells

The channel family is generated by a union of x-, y-, and z-oriented cylindrical fields. The domain-window product selects part of that lattice while the remaining material stays connected.

FIG. 5. Cross-domain field slice

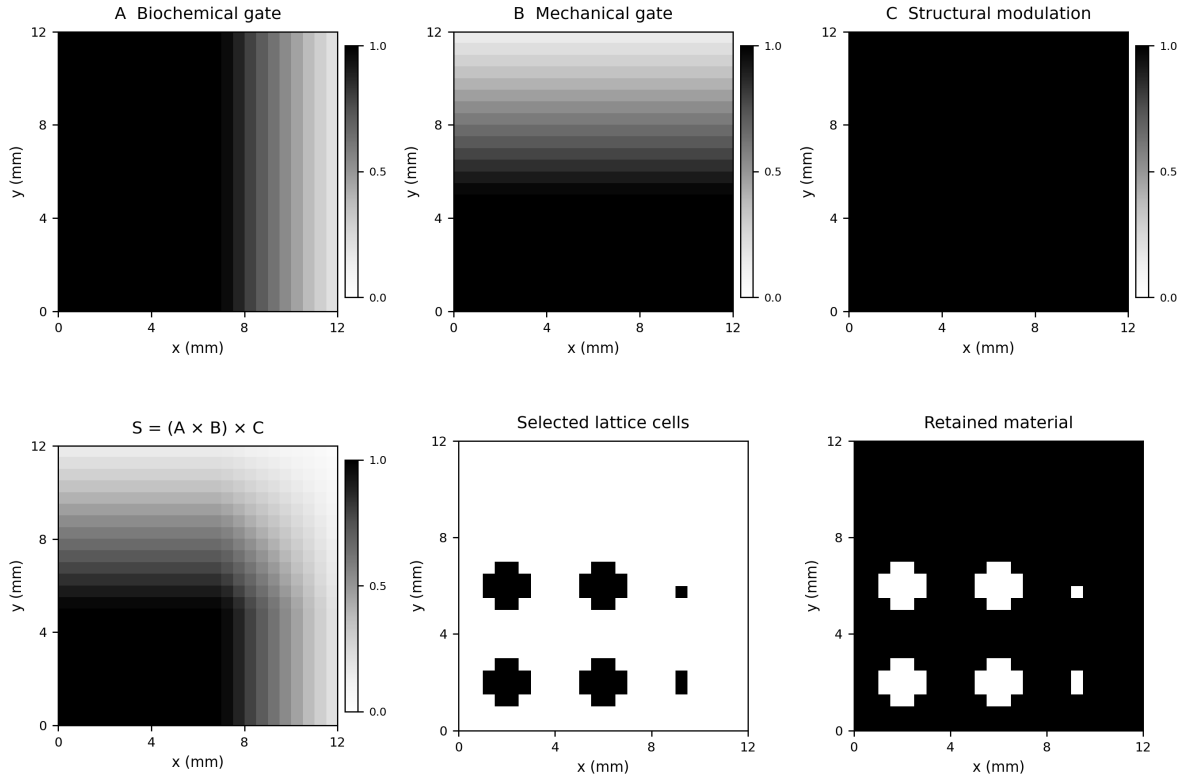


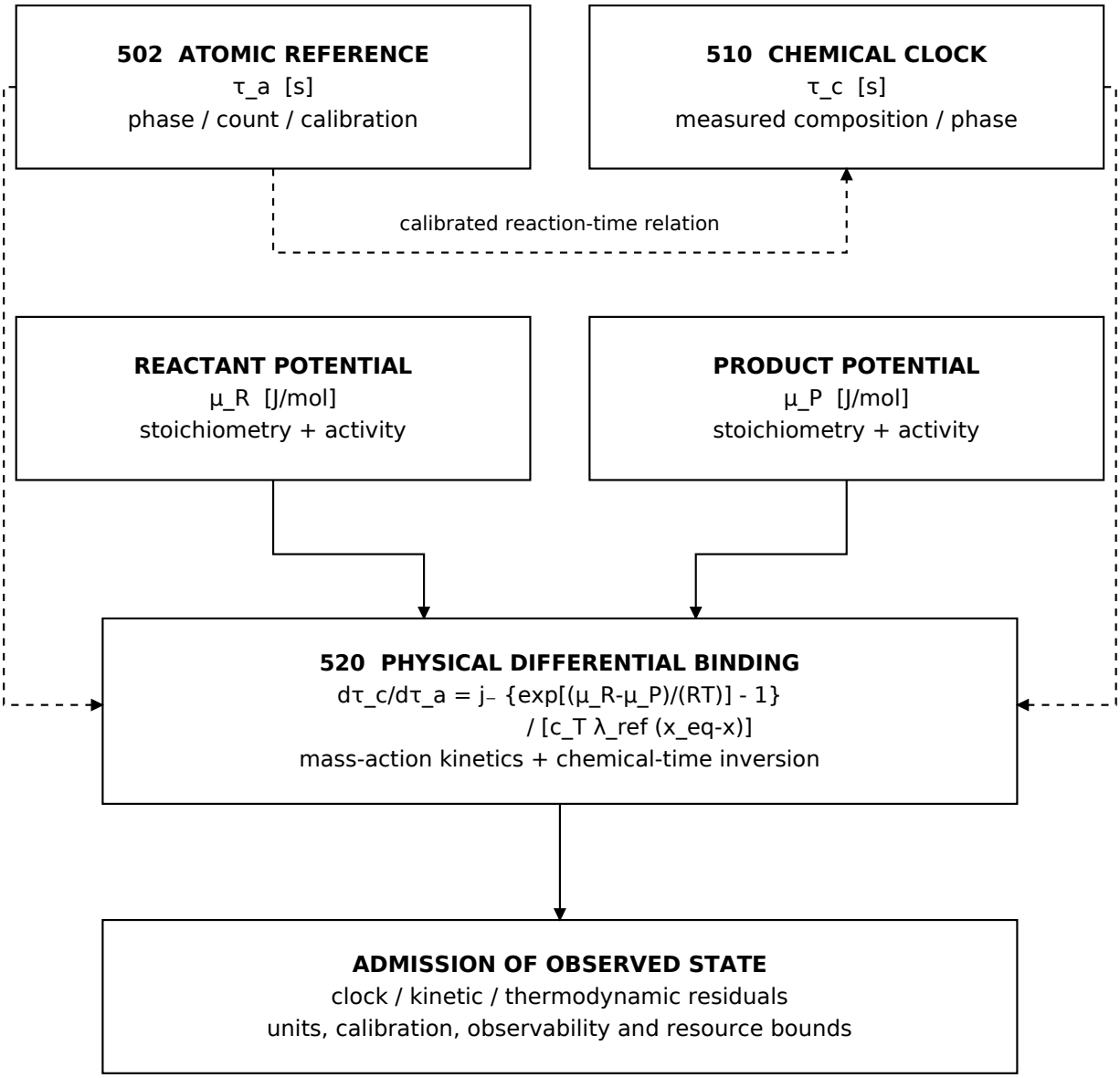
FIG. 5. Common slice k = 8, z = 4.25 mm, at 0.5 mm cell spacing. Darker values in the gate and score panels are larger. In binary panels, black cells indicate selected or retained membership. The selection threshold is 0.55.

LOCAL SELECTION

$D_v = E \text{ AND } V \text{ AND NOT } P \text{ AND } (f_L \leq 0) \text{ AND } (S \geq 0.55)$

$F = E \text{ AND NOT } D_v$

FIG. 6. Physical TAU TAU MU MU binding



TAU TAU MU MU = (τ_a, τ_c, μ_R, μ_P)

FIG. 6. The four-quantity physical binding of Equation 17A. Atomic time and chemically inferred time are connected through the same reaction whose forward/reverse flux ratio is determined by the reactant-to-product chemical-potential difference. Admission uses the declared calibration and observation domain.

FIG. 7. Physical apparatus and feedback

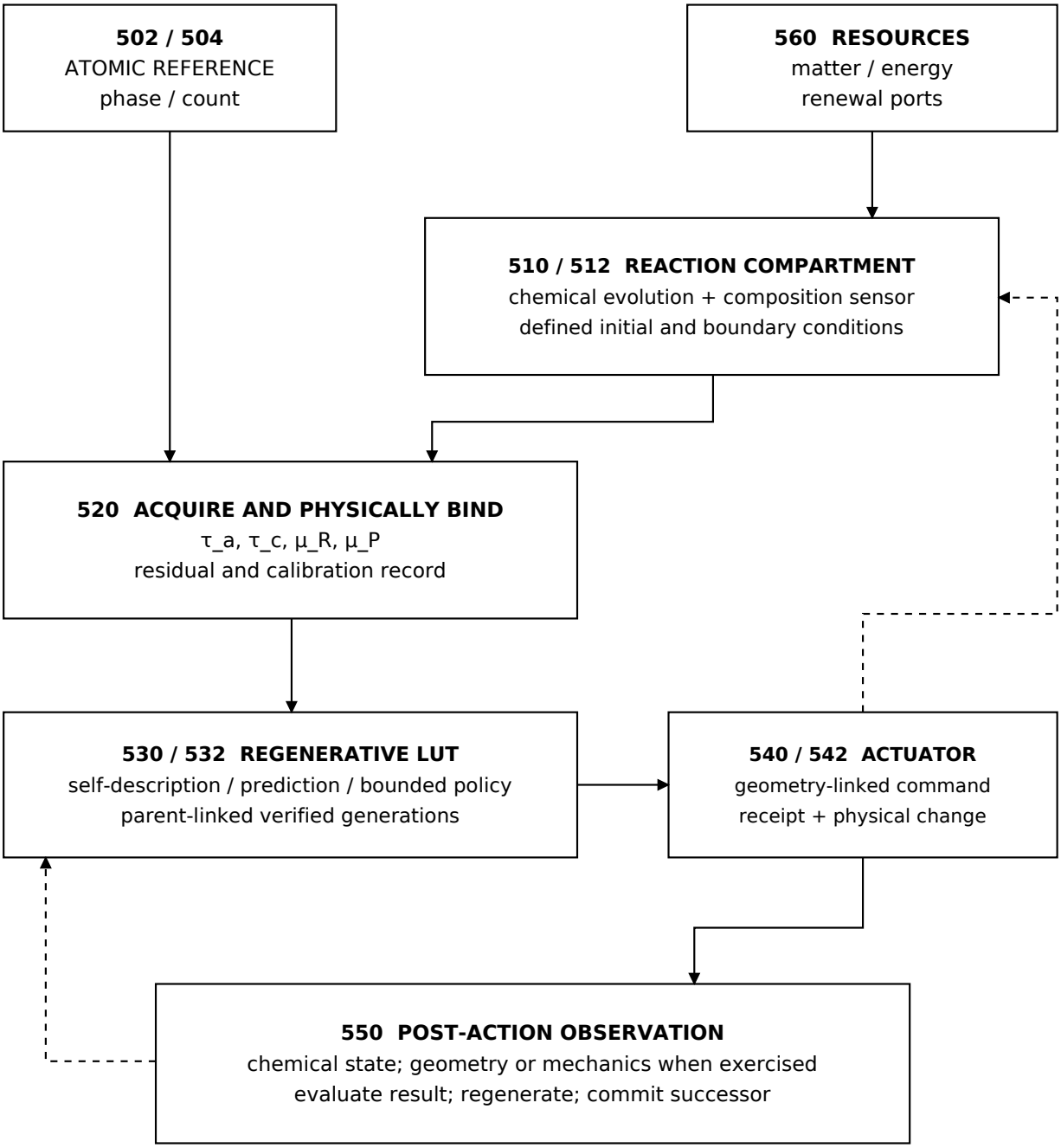
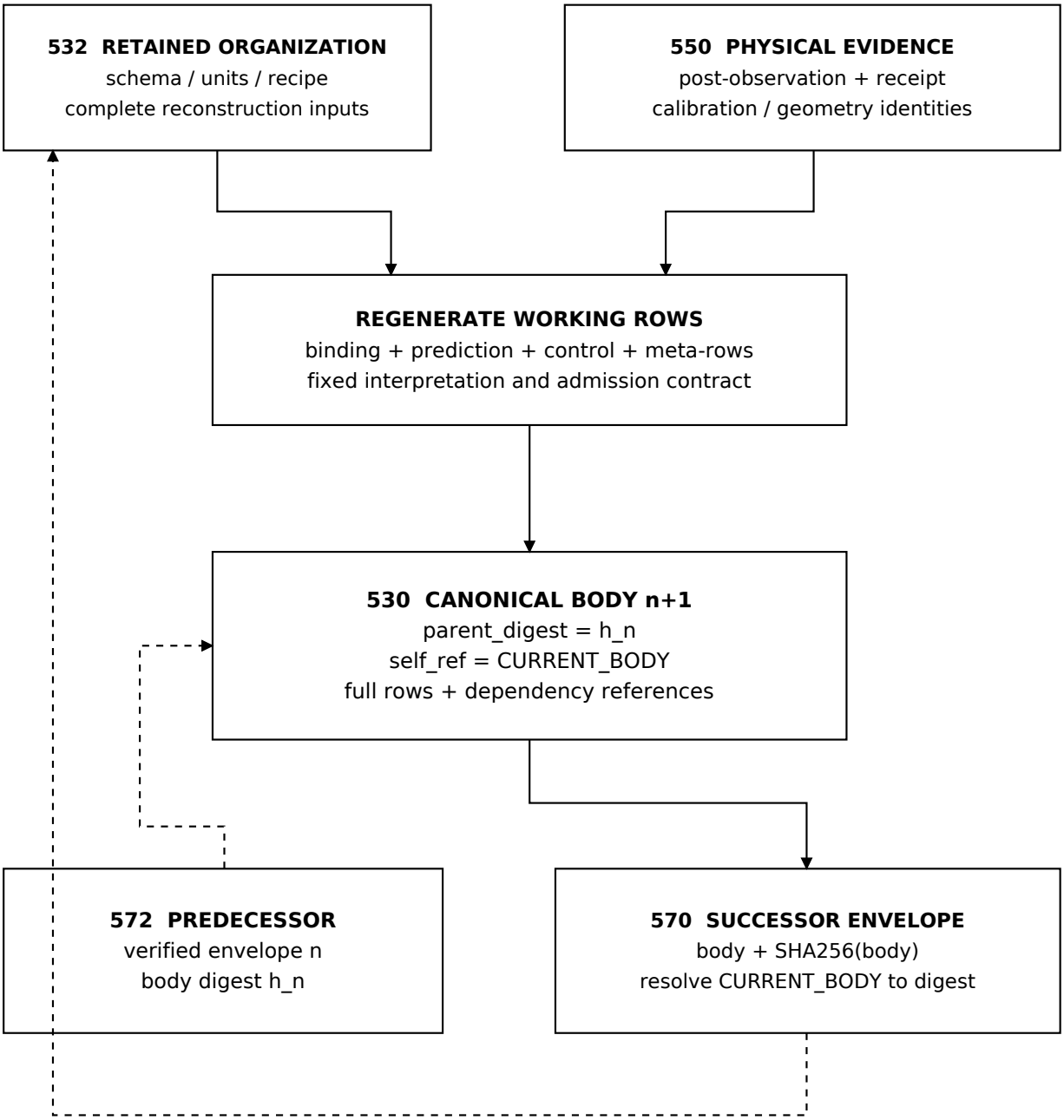


FIG. 7. Apparatus connections between reference timing, reaction state, regenerated control organization, physical actuation and post-action observation. The command mode defines whether the result is a chamber interval, a geometric realization or another registered physical change.

FIG. 8. Self-reference and regeneration



The body excludes its own digest. Full reconstruction data are retained.

FIG. 8. The retained organization and post-action evidence regenerate a canonical table body. Its current digest is held by the envelope; its parent digest identifies the predecessor. Complete snapshots and reconstruction inputs support same-generation rebuilding.

FIG. 9. Atomic and chemical clock binding

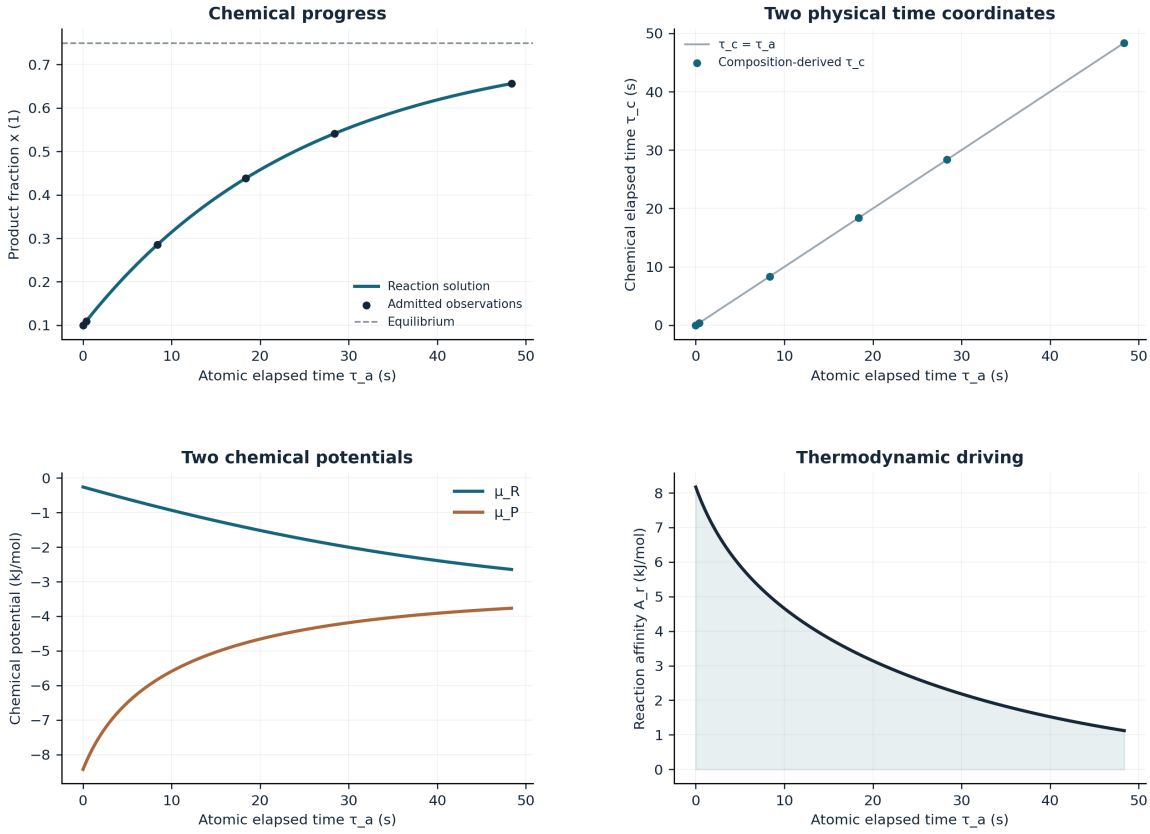


FIG. 9. Analytic-generated R ⇌ P construction at 298.15 K, k_f = 0.03 s^{-1} and k_r = 0.01 s^{-1}. Points show the admitted execution observations. Chemical time is reconstructed from composition. The two potential curves use μ_R° = 0 and μ_P° = -RT ln(3); their difference gives reaction affinity.

FIG. 10. Chemical progression and lattice geometry

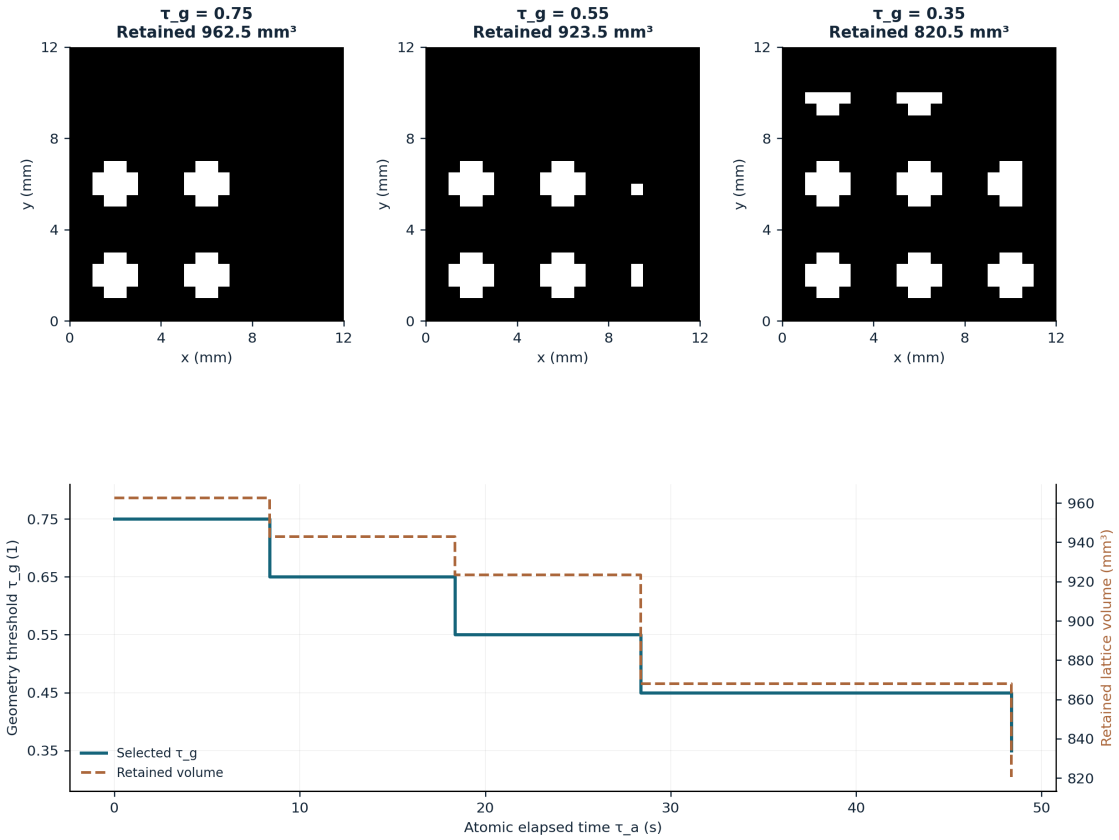


FIG. 10. Three retained-material slices at z = 4.25 mm and a plot of the executed geometry-control schedule. Black denotes retained material. The threshold values are the declared control policy over chemical progression; volume follows the sampled SDF lattice selection.

FIG. 11. Chemical-clock conditioning and model mismatch

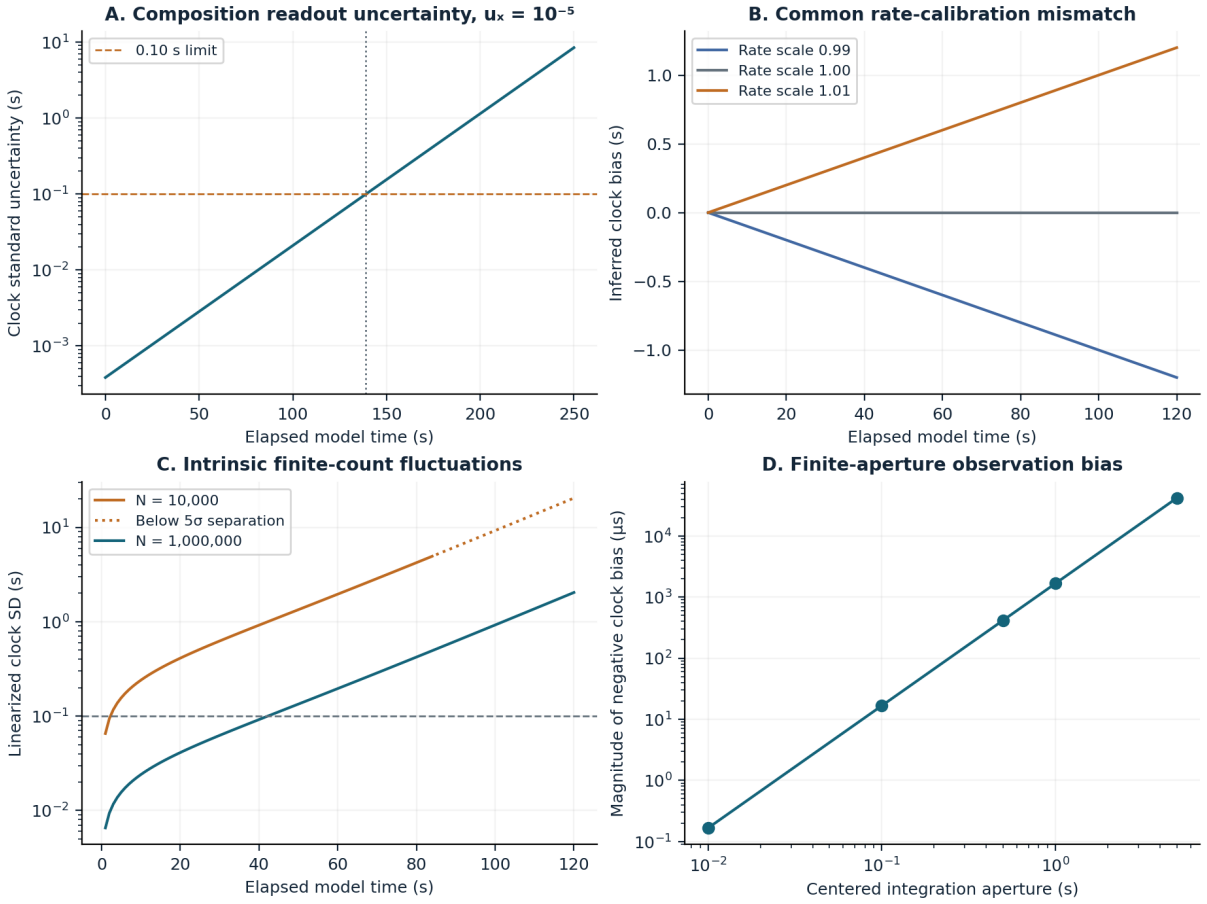


FIG. 11. Synthetic and analytic diagnostics of the stated reference reaction: inverse-clock conditioning, rate-scale mismatch, finite population fluctuations and observation aperture. All curves retain their assumptions and units in the separate audit report.

FIG. 12. Grid resolution and geometric uncertainty

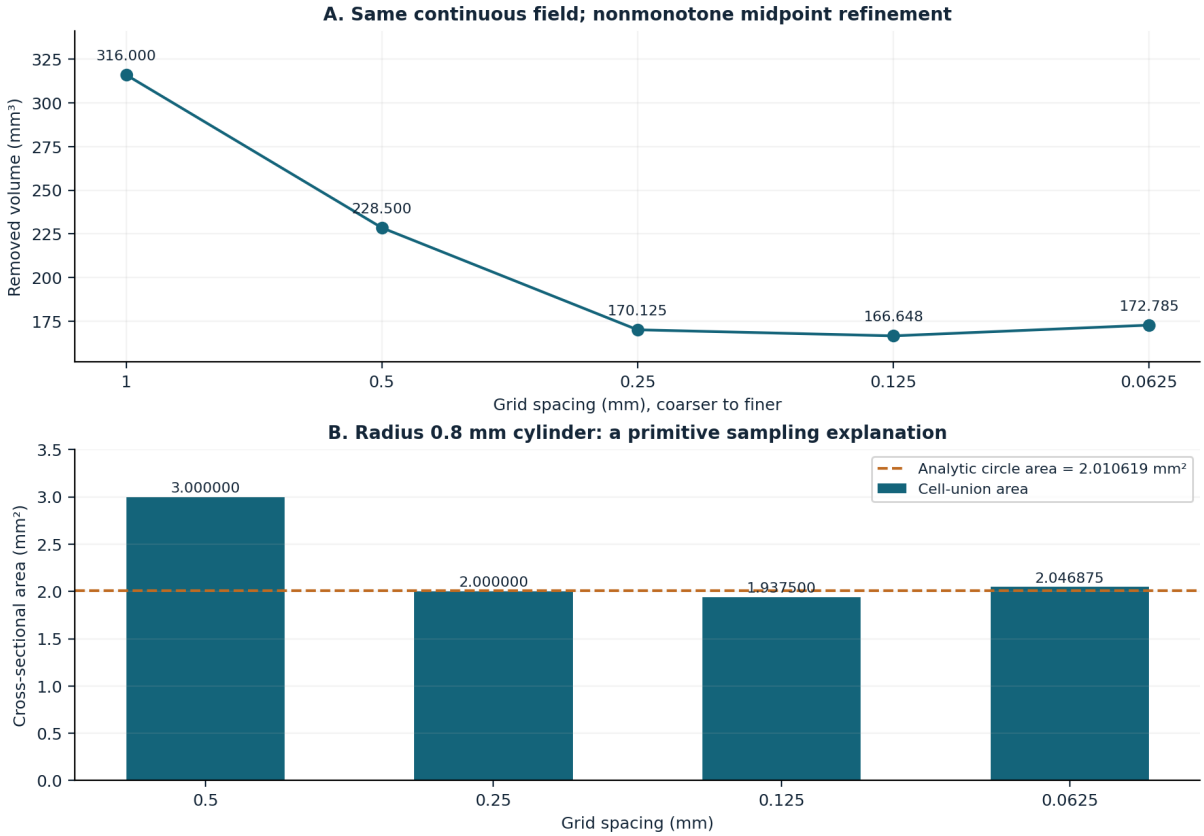


FIG. 12. Grid refinement of the supplied sampled SDF construction and a circular-channel cross-section benchmark. The finest grid is a comparison value, without a certified continuum error bound. The documented threshold and registration perturbation study is retained in the numerical report. Physical fabrication error requires independently observed geometry.

Claim-support map and source record

Claims	Description support
1, 16, 18	Sections 1-7: SDF-window, Hadamard, lattice and CSG construction.
2-6	Sections 2-3: physical predicates, registration, gate maps and exterior collar.
7-13, 17	Sections 5-8: selection, geometric construction, protection, typed output and metadata.
14, 15	Section 8: geometry-driven iteration and physical fabrication.
19, 20	Section 4.3: normalized signed-product and physical-distance bands.
21, 22, 29, 36	Sections 10-15 and 17: integrated physical apparatus and operating method.
23, 24	Sections 10-11, 16 and 17.1: reversible reaction clock and reference compartment.
25, 26, 35	Section 13 and 15: schema regeneration, finite self-reference and complete restoration.
27, 32	Sections 11.3 and 14.3: chemical/component renewal and resource/epoch records.
28, 30, 34	Sections 12, 14 and 17: physical geometry closure, bounded policy and transport coupling.
31, 33	Sections 10.3 and 11: uncertainty, admissible clock segments and oscillator phase.
Physical-closure context	Sections 19-30 state the operating conditions, constitutive choices, uncertainty and empirical evidence needed to realize and evaluate the claimed apparatus. The original 36 claims remain unchanged.

The supplied Markdown specification establishes the biochemical/biomechanical windows, registered array coupling, two Hadamard products, boundary modulation, threshold selection, negative-volume construction, CSG difference, output concept, and original four drawing subjects.

The supplied 14-page discussion PDF records the development of those subjects, including the original prompt, the specification, and the drawing descriptions. The inventor's clarification in this conversation on 20 October 2026 explicitly identifies the signed-distance-field basis and cross-domain lattice and directs supplementation of the missing structures.

The present edition supplies the SDF sign and unit conventions, typed fields, spatial and order-theoretic lattice definitions, operating-set and gate maps, cutter and boundary rules, data records, execution contract, graded embodiments, expanded claims, vector drawings, and reproducible numerical constructions. The package source record retains the uploaded filenames and SHA-256 digests for traceability.

The inventor directs this extension to be autopoietic and self-referential through a LUT, to integrate atomic and chemical clocks, and to carry the name TOWNSENDCAPSTONECORNER. The inventor further clarifies that TAU TAU MU MU binds the formalization to physical equations and processes. This edition implements that instruction as the explicitly defined quartet and physical observation-to-actuation contract in Sections 10-17.

The attached version 1.0 PDF was inspected and its text matched the retained specification source. This edition preserves the geometric disclosure, renames its scalar threshold τ_g to distinguish it from physical clock quantities, and adds the reaction model, physical apparatus, regenerative table, additional claims and integrated execution records. The physical quartet definitions are part of this edition's supplied formalization.

Version 3 responds to the instruction to supply missing equations for physical realization and assess determinism or flaws. It preserves the earlier ideal implementation, adds a separately identified synthetic sensitivity audit, and distinguishes model identities, numerical verification, calibrated measurement and independent physical validation. No new apparatus measurements were supplied for this revision.

Implementation record

The source package contains the numerical inputs, generated meshes, Boolean scripts, clock observations, self-referential table generations, physical-binding residuals, receipt and replay records, tests, editable document, and figure sources. Sections 9 and 16 identify the executed constructions; Sections 10-17 define apparatus embodiments and their observation interfaces. Sections 19-30 add closure equations, quantified sensitivities and the empirical completion register. The separate audit distinguishes numerical verification from physical validation.

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